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# VideoGenDoctor: Auditable Diagnosis and Repair Planning for Controllable Video Generation

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## Abstract

Controllable video generation increasingly depends on character identities, camera trajectories, required props, temporal actions, and multi-shot scripts. Scalar quality or alignment scores help rank models, but they provide limited debugging support because they rarely identify the failure, localize the supporting evidence, name the affected target, or specify a concrete repair.

We present VideoGenDoctor, a diagnosis-and-repair framework that represents generation failures as code–span–target–plan records grounded in temporal evidence and representative keyframes. The default implementation combines lightweight feature-based screening, optional VLM verification, and a patch compiler that maps localized diagnoses to structured repair plans. We evaluate VideoGenDoctor on VideoGenDoctor-Bench-v0, a controlled diagnostic fixture with 324 videos, 9 deterministic perturbation types, and 2,016 human-verified evidence spans covering 12 observed failure codes from a 38-code taxonomy. VideoGenDoctor-diagnosis reaches macro-F1 0.9110 and tIoU@0.5 0.7316; the higher-cost Rule+GPT-4V reference reaches 0.9276 and 0.7688. In closed-loop evaluation, VideoGenDoctor-full improves automatic Pass@1/2 from 24.07%/35.80% for Score-only feedback to 64.81%/83.64%. On a 240-video failure-enriched subset from four controllable video generators, the default pipeline improves diagnosis macro-F1 from 0.7992 to 0.8264 and tIoU@0.5 from 0.5431 to 0.5982 over direct structured VLM reporting. These results support auditable repair records as a practical interface for controlled diagnosis and repair planning, but they do not establish open-world coverage across arbitrary generators, domains, or video lengths.

## 1 Introduction

Recent diffusion-based models can synthesize multi-second videos conditioned on text, reference images, camera trajectories, character identities, and structured shot scripts Yang et al. [2024], Blattmann et al. [2023], Wang et al. [2023], Yuan et al. [2024]. In production-style workflows,

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ShotIR-like specifications define per-shot props, characters, camera movement, shot type, and action order. When a generated video violates these specifications, users need more than a model-level score: they need to know what failed, where the evidence appears, which target is affected, and how the next generation attempt should be changed.

Current video-generation evaluators such as VBench Huang et al. [2024], EvalCrafter Liu et al. [2024], VideoScore He et al. [2024], and T2V-CompBench Sun et al. [2025] report aggregate quality, motion, or alignment scores. These scores are useful for model comparison but weak as debugging signals: a low score does not indicate whether the character identity drifted, a required prop disappeared, camera motion was violated, or a specific temporal segment should be regenerated.

VideoGenDoctor reframes controllable-video evaluation as structured diagnosis and repair planning. Given a video  $V$  and specification  $S$ , it outputs records

$$R = \{(c_i, t_0^i, t_1^i, y_i, \sigma_i, K_i, T_i, a_i)\},$$

where  $c_i$  is a failure code,  $(t_0^i, t_1^i)$  is the evidence span,  $y_i$  is the affected target,  $\sigma_i$  is confidence,  $K_i$  contains representative keyframes,  $T_i$  stores optional track-level evidence, and  $a_i$  is a compiled repair action. The interface is designed to be machine-readable and human-auditable; executable repair depends on the capabilities exposed by the downstream generator or editor.

This framing leads to three empirical questions. First, can structured records improve failure-code detection and temporal localization over direct VLM critiques? Second, do localized evidence and template-constrained plans improve repair utility beyond score-only feedback or code-only patches? Third, how far do conclusions from deterministic perturbations transfer to naturally occurring generator failures without claiming open-world coverage?

**Scope.** We separate the interface claim from the open-world generality claim. The controlled fixture supports the claim that code–span–target–plan records provide auditable diagnosis and useful repair plans under known perturbations. A failure-enriched real-generator subset suggests transfer beyond deterministic perturbations, but broader coverage across generators, domains, styles, and long-form videos remains future work. We therefore call VideoGenDoctor-Bench-v0 a controlled diagnostic fixture rather than a general open-world benchmark.

**Contributions.** This paper contributes: (1) a code–span–target–plan representation for controllable video debugging; (2) a modular framework combining feature-based candidate generation, optional VLM verification, evidence aggregation, and patch compilation; (3) VideoGenDoctor-Bench-v0, with 324 controlled videos, 2,016 verified evidence spans, 9 perturbation types, and 12 observed codes, plus a 240-video failure-enriched real-generator subset covering 18 observed codes; and (4) an evaluation against direct VLM diagnosis, VLM-to-patch repair, blind human repair validation, per-code reliability analysis, paired bootstrap comparisons, and transfer checks on naturally occurring failures.

## 2 Related Work

**Video generation evaluation.** Existing benchmarks and evaluators measure aggregate video quality, motion smoothness, subject consistency, and text-video alignment Huang et al. [2024], Liu et al. [2024], He et al. [2024], Sun et al. [2025]. They are valuable for ranking models but do not usually produce localized evidence or executable repair plans. VideoGenDoctor instead returns structured failure records tied to temporal spans and repair templates. A compact capability comparison is provided in Appendix A, Table 10.

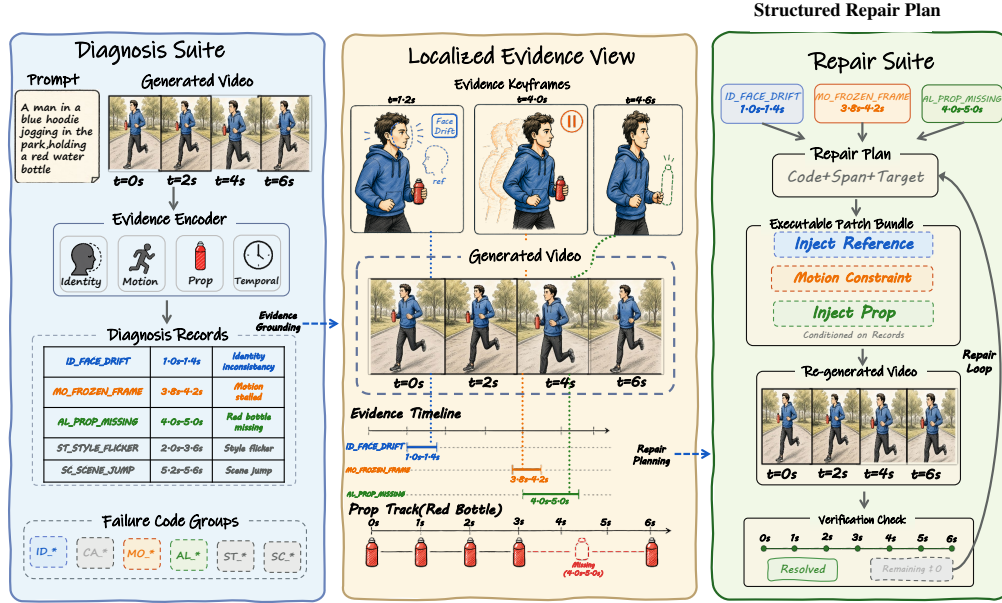
**VLM critics and temporal grounding.** Large vision-language models can inspect sampled frames and produce critiques Liu et al. [2023], Chen et al. [2024], OpenAI [2023]. We therefore treat direct VLM reporting and VLM-to-patch generation as strong baselines rather than strawmen. The central question is whether constraining critique through code–span–target records improves temporal grounding, schema validity, auditability, repair executability, and cost. For temporal localization, we adapt temporal intersection-over-union:

$$\text{tIoU} = \frac{\max(0, \min(\hat{t}_1, t_1^*) - \max(\hat{t}_0, t_0^*))}{\max(\hat{t}_1, t_1^*) - \min(\hat{t}_0, t_0^*)}.$$

74 **Automated repair.** Program repair systems typically localize a defect, classify it, and generate a  
 75 patch. VideoGenDoctor adapts this pattern to controllable video generation: each diagnosis must  
 76 expose evidence and map to a supported repair family, connecting evaluation directly to iterative  
 77 generation.

## 78 3 VideoGenDoctor

### 79 3.1 Overview



**Figure 1:** Overview of VideoGenDoctor. The system converts a generated video and its prompt or structured specification into failure-as-code records, grounds each record in localized temporal evidence, compiles a structured repair plan, and verifies regenerated videos in a bounded closed loop.

80 VideoGenDoctor contains four modules: segmentation and feature extraction, candidate diagnosis,  
 81 optional VLM verification, and patch compilation. The first module samples keyframes and computes  
 82 lightweight signals; the second applies thresholded rules to produce failure candidates; the third  
 83 calibrates top-ranked candidates using structured VLM questions; and the fourth maps retained  
 84 diagnosis records to structured repair actions such as reference injection, prop injection, camera  
 85 override, segment regeneration, or style correction. We distinguish repair-plan validity from adapter  
 86 executability: a plan is valid when it names a supported action, target, and evidence span, while it is  
 87 executable only when the downstream generator or editor exposes the required control interface.

### 88 3.2 Failure Records and Patch Templates

89 VideoGenDoctor uses a machine-readable taxonomy with 38 codes grouped into Identity, Camera,  
 90 Motion, Alignment, Style, and Scene. The current controlled fixture evaluates 12 observed codes; the  
 91 real-generator subset covers additional naturally occurring codes. Codes outside these observed sets  
 92 define an extensible namespace and are not treated as empirically validated. The full taxonomy and  
 93 group-level description are provided in Appendix A and Appendix C.

94 Each predicted failure is represented as

$$r = (c, t_0, t_1, y, \tilde{\sigma}, K, T, a),$$

95 where  $c$  is the taxonomy code,  $(t_0, t_1)$  is the evidence span,  $y$  is the affected target,  $\tilde{\sigma}$  is confidence,  
 96  $K$  stores representative keyframes,  $T$  stores optional tracks, and  $a$  is the compiled repair action.  
 97 Detectors and judges can be replaced without changing this schema.

**Table 1:** Representative mappings from failure types to evidence signals and repair-plan templates. Each diagnosis record stores the failure code, localized span, affected target, confidence, evidence, and compiled repair action.

| Failure type                   | Evidence signal                                    | Repair action                                 |
|--------------------------------|----------------------------------------------------|-----------------------------------------------|
| Face identity drift            | Face embedding drift across keyframes              | Reference-frame injection                     |
| Body / clothing drift          | Character-region appearance drift                  | Reference-frame or character-anchor injection |
| Camera movement deviation      | Camera flow trace or trajectory mismatch           | Camera movement override                      |
| Frozen frames or action stalls | Near-zero flow or stalled pose track               | Regenerate affected segment                   |
| Missing required prop          | Object presence check or VLM prop-absence judgment | Prop injection                                |
| Style inconsistency            | Style or color embedding drift across segments     | Style correction                              |
| Background inconsistency       | Scene/background embedding drift                   | Background anchoring                          |

### 3.3 Candidate Diagnosis and Verification

Given video  $V$ , we partition it into 2-second fixed-stride segments and sample six keyframes per segment. The default implementation computes semantic embedding drift with OpenCLIP Cherti et al. [2023], optical flow with RAFT Teed and Deng [2020] or OpenCV Farneback Bradski [2000], Farneback [2003], face drift with InsightFace Deng et al. [2019], and object or prop cues with YOLOv8 Ultralytics [2023] when enabled. For each segment  $s$  and failure code  $c$ , rule  $\rho_c$  produces a score  $\sigma_{s,c} \in [0, 1]$ ; top-ranked candidates inherit segment boundaries and keyframes ranked by frame-level anomaly scores.

The optional Stage-2 judge receives the top- $K$  candidate segments ( $K = 3$  by default), sampled keyframes, temporal bounds, candidate codes, and relevant prompt or ShotIR fields. It answers structured questions rather than writing an unconstrained report. Final confidence is

$$\tilde{\sigma}_{s,c} = (1 - \alpha)\sigma_{s,c}^{(1)} + \alpha\sigma_{s,c}^{(2)},$$

with  $\alpha = 0.6$  by default. Hosted endpoints, prompts, sampled frames, raw responses, parsed JSON, and latency metadata are logged for reproducibility.

### 3.4 Evidence Localization and Repair Planning

Each retained record stores an evidence object  $E = (t_0, t_1, K, T)$  so that the diagnosis can be audited. Fixed segments are sufficient for the current fixture; future versions can refine spans using frame-level peaks and track-based merging. The patch compiler emits either ShotIR field-level diffs or generator-agnostic repair plans, preserving the triggering code, evidence span, affected target, and rationale. In closed-loop evaluation, a video passes when all failure confidences fall below  $\tau = 0.5$  or the loop reaches  $K_{\max} = 3$ . Because this automatic criterion depends on the verifier, Section 4.5 also reports blind human repair validation.

## 4 Experiments

### 4.1 Benchmark and Evaluation Protocol

We evaluate VideoGenDoctor as a diagnosis-and-repair system rather than as a new video generator. The experiments align with the three questions in the introduction: failure-code correctness, temporal evidence localization, repair-plan usefulness, and final specification satisfaction under independent human judgment.

**Controlled fixture.** VideoGenDoctor-Bench-v0 contains 324 perturbed videos, 2,016 verified evidence spans, 9 deterministic perturbation types, and 12 observed failure codes. It is built from 18 clean source clips across six domain groups; each clip is paired with a ShotIR-style specification and perturbed by nine deterministic operators under two seeds. The fixture validates interface consistency, evidence auditability, and repair-planning usefulness under controlled failures; it is not intended as a

comprehensive open-world benchmark. Construction details and operator-to-template alignment are moved to Appendix A, Table 12, and Appendix B, Table 18.

**Real-failure and real-normal subsets.** We collect 240 naturally occurring failure videos from CogVideoX, Wan, Stable Video Diffusion, and HunyuanVideo Yang et al. [2024], Wan-AI [2025], Blattmann et al. [2023], Tencent Hunyuan Foundation Model Team [2025]. The subset is failure-enriched: from 480 raw generations, we retain videos only when annotators verify at least one visible specification violation. Sampling is balanced across the four generators at 60 retained failure videos each, with generator-specific input adapters logged in a per-video JSON manifest: CogVideoX and Wan use ShotIR-to-prompt text templates, Stable Video Diffusion uses reference-frame conditioning, and HunyuanVideo uses a multi-shot prompt template. The manifest records prompt or spec identifiers, converted inputs, generation settings, screening status, verified codes, and evidence spans so that the transfer check remains auditable even though it is not a prevalence estimate. We also retain 120 real-normal videos from the same raw generations to estimate false positives and unnecessary repair behavior. Ambiguous and low-quality exclusions are treated as stress analyses rather than standard accuracy benchmarks. Detailed source composition, normal-set behavior, temporal evidence profiles, and stress tables are reported in Appendix B.

**Annotation.** A reliability subset of 120 videos and 768 candidate evidence spans is dual-annotated before adjudication. We report code-level Cohen’s  $\kappa$  and span-level annotator-to-annotator tIoU in Appendix B, Table 24. Agreement is substantial for failure codes and lower for exact temporal boundaries, reflecting the ambiguity of fine-grained evidence spans.

**Table 2:** Dataset statistics for the controlled fixture and failure-enriched real-generator subset. The real-failure split contains naturally occurring, annotator-verified specification violations from multiple generators and is not used to estimate failure prevalence.

| Split               | #Vid | #Evidence | AvgDur | PerturbTypes | #Codes |
|---------------------|------|-----------|--------|--------------|--------|
| Controlled fixture  | 324  | 2016      | 10.46s | 9            | 12     |
| Real-failure subset | 240  | 1536      | 8.93s  | –            | 18     |

## 4.2 Compared Methods and Metrics

We compare internal diagnosis variants (Prior-only, Rule-only, Rule+DummyJudge, Rule+Open-VLM, Rule+GPT-4V, and VideoGenDoctor-diagnosis), repair variants (VideoGenDoctor-patch, Patch+Judge, and VideoGenDoctor-full), and external VLM baselines including free-form VLM reporting, structured VLM reporting, self-consistency, VLM temporal proposal + patch, VLM-to-patch, and Score+LLM-to-patch. Most direct VLM baselines share the same hosted GPT-4V endpoint as Rule+GPT-4V to isolate prompt structure, output schema, and repair-planning protocol rather than model backbone; Rule+Open-VLM is included as an open-model reference. Whenever a table reports human repair success for a diagnosis-oriented front end, that front end is paired with the same downstream repair loop so that only the diagnostic interface changes. Implementation details are reported in Appendix B, Table 25; strengthened VLM-agent results are summarized in Appendix A, Table 13.

We report macro-F1 and micro-F1 for failure-code detection, tIoU@0.3/0.5 and Top- $K$  keyframe hit rate for localization, automatic and human Pass@1/2 for repair, patch usefulness, new-artifact rate, and video-level bootstrap 95% confidence intervals. All span-aware metrics use the same matching protocol: records are grouped by failure code, target-incompatible predictions are filtered, and remaining pairs are matched by Hungarian assignment Kuhn [1955] using tIoU as the primary score and confidence only as a tie-breaker. Invalid or missing spans receive no localization credit. For paired repair comparisons, we report bootstrap differences and treat small point estimates as inconclusive when the confidence interval crosses zero.

### 4.3 Failure-Code Detection

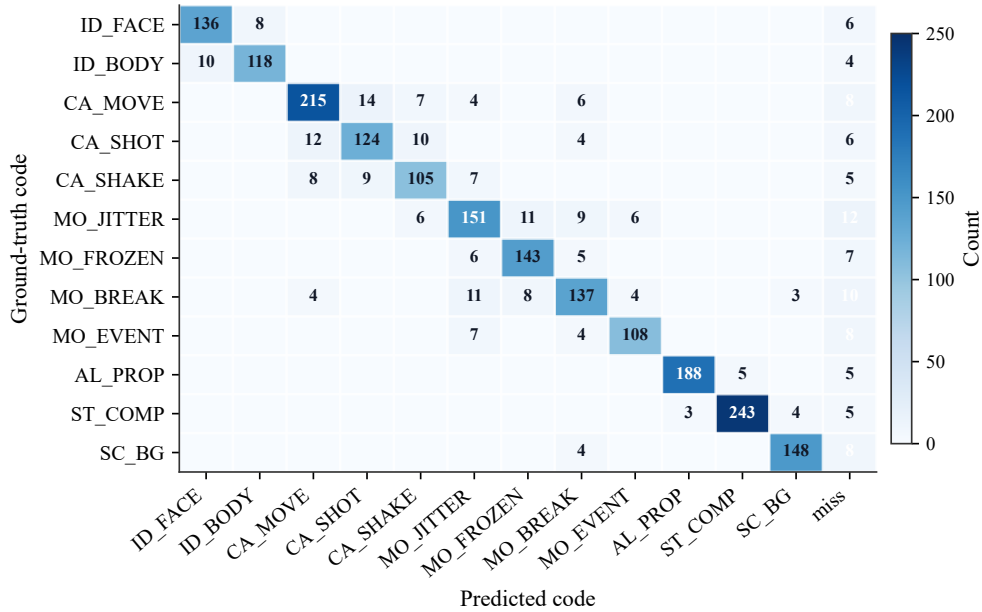
**Table 3:** Failure-code detection on the controlled fixture. Direct VLM baselines test whether schema-constrained diagnosis improves over strong free-form and structured critique alternatives. Confidence intervals are estimated by video-level bootstrap resampling; best values are boldfaced.

| Method                   | Macro-F1      | Micro-F1      | 95% CI         |
|--------------------------|---------------|---------------|----------------|
| Prior-only               | 0.4479        | 0.4635        | [0.419, 0.477] |
| Rule+DummyJudge          | 0.8584        | 0.8716        | [0.836, 0.879] |
| Rule-only                | 0.8926        | 0.9038        | [0.872, 0.911] |
| Rule+Open-VLM            | 0.9047        | 0.9175        | [0.885, 0.922] |
| VideoGenDoctor-diagnosis | 0.9110        | 0.9243        | [0.893, 0.928] |
| Rule+GPT-4V              | <b>0.9276</b> | <b>0.9386</b> | [0.912, 0.941] |
| VLM free-form report     | 0.8402        | 0.8627        | [0.814, 0.866] |
| VLM structured report    | 0.8869        | 0.9018        | [0.864, 0.908] |

VideoGenDoctor-diagnosis outperforms free-form and structured direct VLM reporting on macro-F1, while Rule+GPT-4V remains the strongest configuration. The stronger localization results below suggest that the benefit of structured diagnosis is not limited to detecting a failure; the records also ground failures stably enough to support repair planning.

### 4.4 Per-Code Reliability and Evidence Localization

The controlled fixture covers twelve observed codes. Detailed per-code precision, recall, F1, and tIoU are reported in Appendix B, Table 26; the confusion matrix below summarizes dominant error modes.



**Figure 2:** Per-code confusion matrix for the 12 observed taxonomy codes on the controlled fixture. The strongest off-diagonal confusions occur among camera and motion failures, while identity, prop-missing, compression-artifact, and scene-background failures remain comparatively concentrated on the diagonal.

**Table 5:** Closed-loop repair evaluation on the controlled fixture. Automatic metrics are computed on all 324 controlled videos. Human metrics are computed on a stratified blind-evaluation subset of 144 videos. Best pass rates and patch-usefulness values are boldfaced; the new-artifact rate is reported for context rather than bold-ranked because abstention-heavy baselines are not directly comparable.

| Method              | Auto P@1      | 95% CI         | Auto P@2      | 95% CI         | Human Pass@1  | 95% CI         | Human Pass@2  | 95% CI         | Patch useful | New artifacts |
|---------------------|---------------|----------------|---------------|----------------|---------------|----------------|---------------|----------------|--------------|---------------|
| Score-only          | 0.2407        | [0.196, 0.291] | 0.3580        | [0.307, 0.411] | 0.1875        | [0.130, 0.257] | 0.2778        | [0.207, 0.355] | 2.28         | 0.0833        |
| Patch-only          | 0.5247        | [0.471, 0.579] | 0.6975        | [0.646, 0.744] | 0.4792        | [0.399, 0.559] | 0.6319        | [0.551, 0.706] | 3.50         | 0.1736        |
| VLM-to-patch        | 0.5617        | [0.507, 0.615] | 0.7438        | [0.694, 0.788] | 0.5069        | [0.427, 0.586] | 0.6736        | [0.594, 0.746] | 3.64         | 0.2153        |
| Patch+Judge         | 0.6296        | [0.575, 0.681] | 0.8179        | [0.773, 0.857] | 0.5625        | [0.482, 0.640] | 0.7431        | [0.667, 0.807] | 3.93         | 0.1458        |
| VideoGenDoctor-full | <b>0.6481</b> | [0.594, 0.699] | <b>0.8364</b> | [0.793, 0.873] | <b>0.5833</b> | [0.503, 0.660] | <b>0.7639</b> | [0.689, 0.826] | <b>4.11</b>  | 0.1319        |

**Table 4:** Evidence localization against human evidence spans on the controlled fixture. Confidence intervals are estimated by video-level bootstrap resampling; best values are boldfaced.

| Method                   | tIoU@0.3      | tIoU@0.5      | Top-1         | Top-3         | 95% CI         |
|--------------------------|---------------|---------------|---------------|---------------|----------------|
| Prior-only               | 0.5817        | 0.4295        | 0.1188        | 0.3315        | [0.403, 0.457] |
| Rule+DummyJudge          | 0.7282        | 0.6128        | 0.2241        | 0.4259        | [0.586, 0.639] |
| Rule-only                | 0.7836        | 0.6714        | 0.2670        | 0.5123        | [0.645, 0.698] |
| Rule+Open-VLM            | 0.8089        | 0.7062        | 0.3565        | 0.5941        | [0.681, 0.731] |
| VideoGenDoctor-diagnosis | 0.8261        | 0.7316        | 0.4336        | 0.6815        | [0.707, 0.754] |
| Rule+GPT-4V              | <b>0.8554</b> | <b>0.7688</b> | <b>0.4923</b> | <b>0.7747</b> | [0.746, 0.790] |
| VLM free-form report     | 0.7204        | 0.5986        | 0.2994        | 0.5117        | [0.570, 0.625] |
| VLM structured report    | 0.7812        | 0.6639        | 0.3651        | 0.5858        | [0.637, 0.690] |

#### 4.5 Closed-Loop Repair and Human Validation

Closed-loop repair is evaluated on the controlled fixture. Automatic Pass@1/2 is computed on all 324 videos using the system verifier, while blind human Pass@1/2 is computed on a stratified 144-video subset. Human raters see the specification and output video but not method names or diagnosis records, so the human metric is the primary check against verifier-coupled gains.

The main repair gain is most consistent with structured repair planning and verification rather than with the final loop wrapper alone. The paired comparison in Section 4.7 shows that the difference between Patch+Judge and VideoGenDoctor-full is small and not statistically decisive, so the full loop is best interpreted as traceable workflow integration.

#### 4.6 Controlled Fixture versus Real Generator Failures

**Table 6:** Transfer from controlled perturbations to naturally occurring generator failures. Macro-F1 and tIoU@0.5 evaluate each diagnosis front end; patch usefulness and Human Pass@2 evaluate the corresponding downstream repair protocol. Best values within each subset are boldfaced.

| Subset              | Method                              | Macro-F1      | tIoU@0.5      | Patch useful | Human Pass@2  |
|---------------------|-------------------------------------|---------------|---------------|--------------|---------------|
| Controlled fixture  | VLM structured report + repair loop | 0.8869        | 0.6639        | 3.62         | 0.6810        |
| Controlled fixture  | VideoGenDoctor default pipeline     | <b>0.9110</b> | <b>0.7316</b> | <b>4.11</b>  | <b>0.7639</b> |
| Real-failure subset | VLM structured report + repair loop | 0.7992        | 0.5431        | 3.36         | 0.6075        |
| Real-failure subset | VideoGenDoctor default pipeline     | <b>0.8264</b> | <b>0.5982</b> | <b>3.82</b>  | <b>0.6908</b> |

All methods degrade on naturally occurring failures, confirming that the controlled fixture is not a substitute for open-world evaluation. The real-failure subset contains more overlapping artifacts, less isolated evidence, and weaker alignment between specification fields and visible deviations. Because this subset is failure-enriched and balanced by construction, these numbers measure transfer under verified failures rather than real-world prevalence. Per-code and per-generator results are reported in Appendix B, Tables 27 and 28.

**Table 7:** Repair ablation on the controlled fixture. The variants isolate the contributions of failure codes, affected targets, temporal spans, evidence keyframes, patch templates, and verification. Best utility values are boldfaced; the new-artifact rate is reported for context rather than bold-ranked.

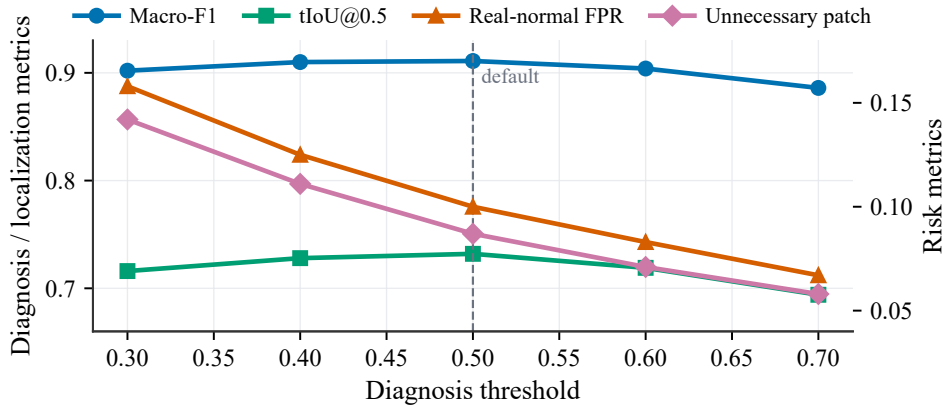
| Repair setting           | Code     | Target   | Span     | Evidence | Template | Human Pass@2  | Patch useful | New artifacts |
|--------------------------|----------|----------|----------|----------|----------|---------------|--------------|---------------|
| Score-only               | ✗        | ✗        | ✗        | ✗        | ✗        | 0.2778        | 2.28         | 0.0833        |
| Code-only patch          | ✓        | ✗        | ✗        | ✗        | ✓        | 0.4583        | 2.91         | 0.1597        |
| Code+target patch        | ✓        | ✓        | ✗        | ✗        | ✓        | 0.5417        | 3.24         | 0.1736        |
| Code+span patch          | ✓        | ✗        | ✓        | ✗        | ✓        | 0.6250        | 3.57         | 0.1667        |
| Code+span+evidence patch | ✓        | ✗        | ✓        | ✓        | ✓        | 0.6875        | 3.80         | 0.1528        |
| VLM-to-patch             | optional | optional | optional | optional | ✗        | 0.6736        | 3.64         | 0.2153        |
| Patch+Judge              | ✓        | ✓        | ✓        | ✓        | ✓        | 0.7431        | 3.93         | 0.1458        |
| VideoGenDoctor-full      | ✓        | ✓        | ✓        | ✓        | ✓        | <b>0.7639</b> | <b>4.11</b>  | 0.1319        |

The ablation tests whether localized evidence and template-constrained repair contribute beyond code prediction alone under a shared backend. The improvement from code-only to code+span and code+span+evidence settings supports the claim that evidence localization is operationally useful for repair planning, but it does not by itself identify a single causal component because target fields, span quality, and backend executability interact in the repair loop.

**Threshold sensitivity.** We further examine the effect of the diagnosis threshold used to retain failure records before repair planning. Table 8 and Figure 3 summarize a draft threshold sweep over  $\tau \in \{0.30, 0.40, 0.50, 0.60, 0.70\}$ . The default threshold  $\tau = 0.50$  provides the strongest balance in this sweep: macro-F1 and tIoU@0.5 peak near the default setting, while real-normal false positives and unnecessary patching decrease monotonically as the threshold becomes more conservative. The trend supports using an intermediate threshold rather than an aggressive low-threshold setting that over-triggers repair plans or a high-threshold setting that suppresses useful diagnoses.

**Table 8:** Threshold sensitivity for retaining diagnosis records before repair planning. Lower thresholds increase repair coverage but also raise real-normal false positives and unnecessary patching; higher thresholds reduce repair risk at the cost of missed diagnoses.

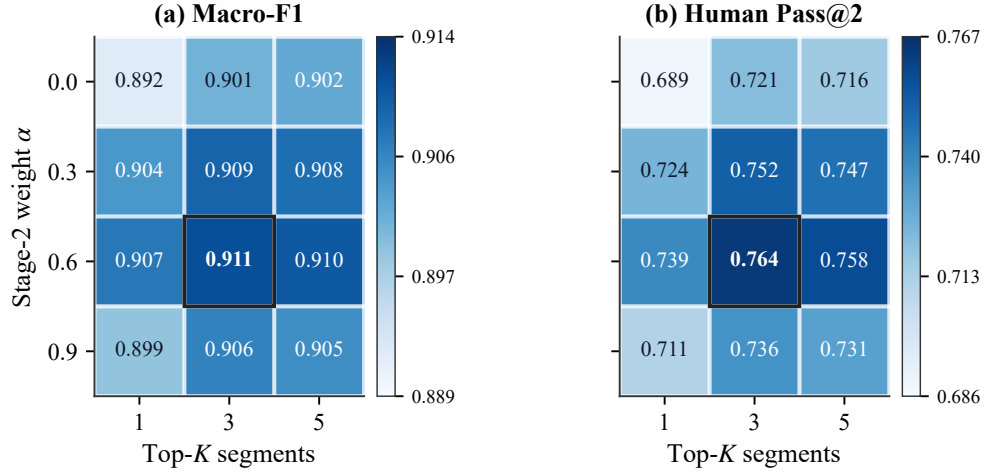
| Threshold $\tau$ | Macro-F1     | tIoU@0.5     | Real-normal FPR | Unnecessary patch rate |
|------------------|--------------|--------------|-----------------|------------------------|
| 0.30             | 0.902        | 0.716        | 0.158           | 0.142                  |
| 0.40             | 0.910        | 0.728        | 0.125           | 0.111                  |
| 0.50             | <b>0.911</b> | <b>0.732</b> | 0.100           | 0.087                  |
| 0.60             | 0.904        | 0.719        | 0.083           | 0.071                  |
| 0.70             | 0.886        | 0.694        | <b>0.067</b>    | <b>0.058</b>           |



**Figure 3:** Threshold sensitivity of diagnosis and repair-risk metrics. The default threshold  $\tau = 0.50$  is marked with a vertical dashed line and gives the best diagnosis-localization trade-off, while more conservative thresholds reduce false positives and unnecessary patching.



**Stage-2 weight and candidate budget.** We also sweep the Stage-2 judge weight  $\alpha$  and the number of candidate segments passed to the judge. Figure 4 shows that  $\alpha = 0.6$  and top- $K = 3$  provide the best operating point among the tested settings. Removing the Stage-2 contribution ( $\alpha = 0$ ) weakens both macro-F1 and Human Pass@2, while assigning too much weight to the judge ( $\alpha = 0.9$ ) reduces stability, consistent with occasional VLM over-correction. Increasing top- $K$  from 1 to 3 improves performance, but moving from 3 to 5 gives little additional benefit and slightly increases exposure to noisy candidates.



**Figure 4:** Sensitivity to Stage-2 judge weight  $\alpha$  and top- $K$  candidate segments. Each cell reports the draft metric value for a particular  $(\alpha, K)$  pair. The default setting,  $\alpha = 0.6$  and  $K = 3$ , yields the strongest balance between diagnosis accuracy and downstream human repair success.

**Table 9:** Paired comparison of repair variants. Differences are paired bootstrap estimates in blind human Pass@2. Small differences between Patch+Judge and the full loop should be interpreted cautiously.

| Comparison                         | $\Delta$ Human Pass@2 | 95% CI          | Interpretation       |
|------------------------------------|-----------------------|-----------------|----------------------|
| Patch-only vs Score-only           | 0.3541                | [0.250, 0.458]  | substantial gain     |
| VLM-to-patch vs Patch-only         | 0.0417                | [-0.054, 0.139] | small / not decisive |
| Patch+Judge vs Patch-only          | 0.1112                | [0.021, 0.215]  | verification helps   |
| VideoGenDoctor-full vs Patch+Judge | 0.0208                | [-0.063, 0.104] | small / not decisive |

We also perform leave-one-perturbation-out and leave-one-source-group-out evaluations. Performance drops under held-out perturbations, especially for missing-event, camera-shake, and temporal-discontinuity operators, but remains above prior-only and score-only baselines. These results suggest that the system is not merely memorizing perturbation templates, while also confirming that unseen-operator generalization is weaker than in-distribution controlled evaluation. Additional draft checks cover adapter executability, stronger real-failure stress slices, and multi-annotator stability. Detailed tables and figures are reported in Appendix B, Tables 29, 30, 15, 16, and 17.

## 5 Discussion

The current evidence supports a scoped claim: structured code–span–target–plan records improve diagnosis and repair planning over score-only feedback and direct VLM reporting on the controlled fixture and the current failure-enriched real-generator subset. The evidence does not show that VideoGenDoctor solves open-world video repair. The controlled fixture demonstrates auditable localization under known perturbations, while the real-failure subset provides an initial but still limited transfer check.

The value of VideoGenDoctor should be assessed as an interface contribution rather than as a new video generation model or a learned repair algorithm. Its advantages lie in schema validity, temporal

230 grounding, repair-plan usefulness, human auditability, and cost-performance. Rule+GPT-4V + repair  
231 loop is the strongest absolute scorer in the current controlled evaluation, but it relies on a more  
232 expensive hosted front end and is best interpreted as an upper-reference configuration rather than  
233 the default operating point. We therefore headline VideoGenDoctor-full because it preserves the  
234 structured interface and audit trail while delivering the main closed-loop gain at the paper’s default  
235 cost point. The modular design also allows feature extractors, rules, VLM judges, and patch compilers  
236 to be replaced independently; Stage-2 judging is restricted to top-ranked candidate segments, and the  
237 closed loop bounds repair by  $K_{\max}$ .

238 **Limitations.** The fixture remains controlled; the real-failure subset is failure-enriched and cannot  
239 estimate natural failure prevalence; automatic pass rates are partly verifier-dependent; only observed  
240 taxonomy codes support empirical claims; and repair assumes access to a generator or editor that  
241 can consume at least part of the patch vocabulary. The current reliability study also uses adjudicated  
242 two-annotator labels, so broader validation requires larger generator coverage, longer videos, more  
243 domains and styles, stronger adapter executability tests, and larger multi-annotator studies.

## 244 6 Conclusion

245 We presented VideoGenDoctor, a structured framework for failure diagnosis, evidence localization,  
246 and repair planning in controllable video generation. The core contribution is an auditable code–span–  
247 target–plan interface that connects evaluation to actionable repair planning. The evaluation combines  
248 a 324-video controlled diagnostic fixture with 2,016 verified evidence spans and a 240-video failure-  
249 enriched real-generator subset with 1,536 verified evidence spans. The controlled fixture covers  
250 12 observed taxonomy codes under 9 deterministic perturbation operators, while the real-generator  
251 subset covers 18 observed taxonomy codes across four publicly identifiable video generation systems.  
252 The main empirical gain comes from structured repair planning and independent verification; the full  
253 closed loop provides traceable integration from diagnosis to evidence, patch planning, re-rendering,  
254 and verification. The broader generality claim remains bounded by the controlled and failure-enriched  
255 nature of the current evaluation sets.

## 256 7 Ethics Statement

257 VideoGenDoctor is a diagnostic tool intended to improve the quality and controllability of generated  
258 videos. Any tool that improves generation reliability could also be misused to strengthen misleading  
259 media. In sensitive deployments, automated repair should be paired with provenance tracking,  
260 watermarking, human review, and access control. Evidence localization may surface keyframes  
261 containing sensitive information, so downstream systems should support local processing, keyframe  
262 redaction, coarse time-binning, and privacy-preserving aggregation where appropriate. The taxonomy  
263 also encodes normative judgments about what counts as a failure; creative workflows should expose  
264 thresholds and permit human override.

## 265 8 Reproducibility and Release

266 For artifact review, the planned release package includes the reference implementation, report  
267 schema, taxonomy, patch compiler, controlled fixture generation scripts, annotations, prediction logs,  
268 evaluation scripts, prompt templates for VLM baselines, and scripts for bootstrap confidence intervals.  
269 Each run is expected to record configuration files, model versions, package versions, random seeds,  
270 prompts, raw VLM responses, and output paths. The intended artifact should permit CPU-only  
271 recomputation of tables from serialized predictions and annotations in seconds; full feature extraction  
272 and VLM judging remain configuration-dependent.

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## 313 A Supplementary Main-Text Results

314 This appendix stores tables and figures moved out of the main text to keep the core narrative compact  
315 while preserving the audit trail for comparisons, taxonomy scope, fixture construction, temporal  
316 evidence profiles, strengthened VLM baselines, repair visualization, and cost-performance trade-offs.

**Table 10:** Capability comparison with related evaluation and critique paradigms. Here “Global score” denotes scalar quality, alignment, or verifier-style pass outputs rather than localized evidence records. Direct VLM reporting is treated as a strong baseline rather than only comparing against score-only benchmarks.

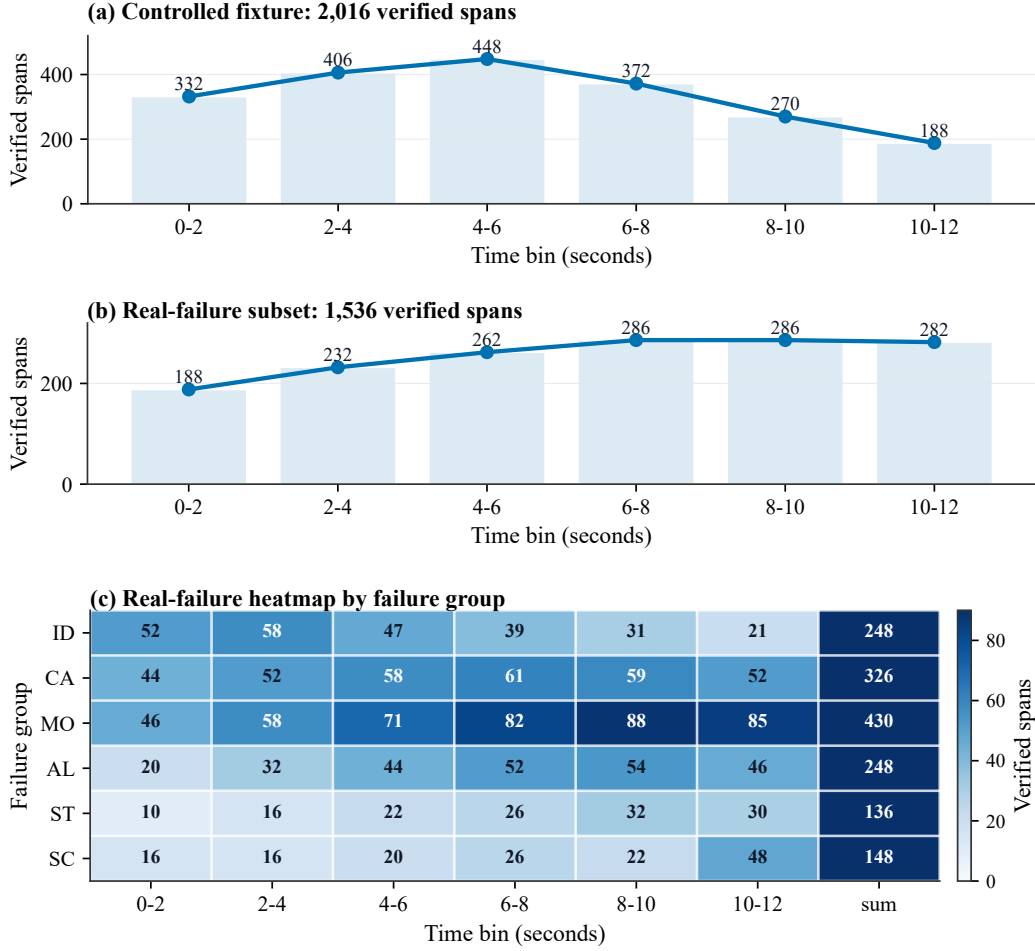
| System                | Global score | Code    | Evidence | Patch   | Closed-Loop |
|-----------------------|--------------|---------|----------|---------|-------------|
| VBench                | ✓            | ✗       | ✗        | ✗       | ✗           |
| EvalCrafter           | ✓            | ✗       | ✗        | ✗       | ✗           |
| VideoScore            | ✓            | ✗       | ✗        | ✗       | ✗           |
| T2V-CompBench         | ✓            | partial | ✗        | ✗       | ✗           |
| Direct VLM report     | partial      | partial | partial  | partial | ✗           |
| <b>VideoGenDoctor</b> | ✓            | ✓       | ✓        | ✓       | ✓           |

**Table 11:** Failure-as-code taxonomy groups. Each group defines a namespace for diagnosis records and links to evidence procedures and repair-plan templates.

| Group     | Operational scope                                                                 |
|-----------|-----------------------------------------------------------------------------------|
| Identity  | Identity drift, face/body consistency, wrong character, clothing consistency.     |
| Camera    | Shot type, camera movement, viewpoint angle, unintended zoom, camera shake.       |
| Motion    | Action timing, frame continuity, motion physics, frozen frames, segment breaks.   |
| Alignment | Prompt adherence, required objects and props, character count, prop placement.    |
| Style     | Color consistency, texture flicker, compression artifacts, art-style consistency. |
| Scene     | Background consistency, lighting, environment match, scene-object stability.      |

**Table 12:** Composition of the controlled diagnostic fixture. Each domain group contains three clean source clips. Every source clip is paired with a ShotIR-style specification and perturbed by nine deterministic operators under two random seeds.

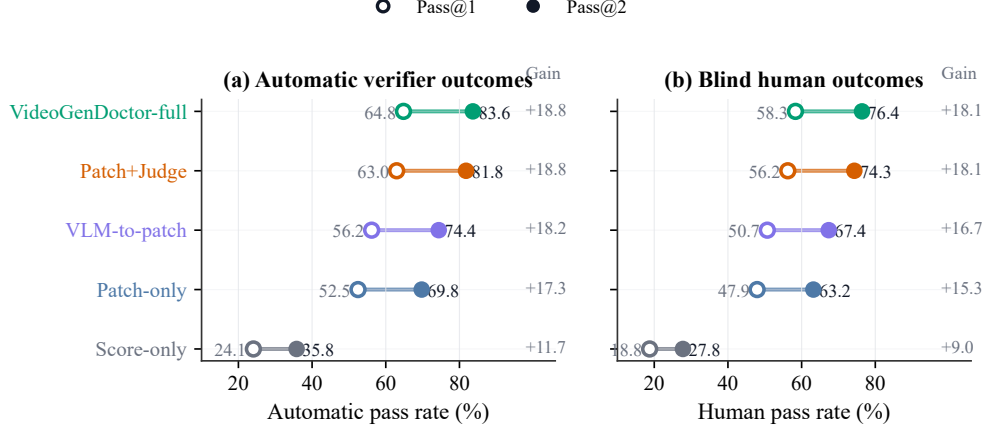
| Domain group               | #Source clips | Avg. duration | Resolution | #Perturb. ops | #Seeds | #Videos |
|----------------------------|---------------|---------------|------------|---------------|--------|---------|
| Indoor / human-object      | 3             | 10.1s         | 768×432    | 9             | 2      | 54      |
| Outdoor / motion           | 3             | 10.8s         | 768×432    | 9             | 2      | 54      |
| Multi-object interaction   | 3             | 10.4s         | 768×432    | 9             | 2      | 54      |
| Camera-control scene       | 3             | 11.0s         | 768×432    | 9             | 2      | 54      |
| Style / lighting variation | 3             | 10.2s         | 768×432    | 9             | 2      | 54      |
| Scene-change / background  | 3             | 10.3s         | 768×432    | 9             | 2      | 54      |
| Total                      | 18            | 10.46s        | –          | 9             | 2      | 324     |



**Figure 5:** Temporal distribution of verified evidence spans in the evaluation sets. Panel (a) aggregates the 2,016 controlled-fixture spans over 2-second bins and shows the front- and mid-loaded pattern expected from deterministic perturbation operators. Panel (b) summarizes the 1,536 real-failure spans over the same time bins, making the later and more diffuse temporal mass directly comparable to panel (a). Panel (c) further breaks the real-failure spans down by failure group and time bin, highlighting that motion and camera failures dominate the later intervals and overlap more strongly than in the controlled fixture.

**Table 13:** Strengthened VLM-agent baselines on the controlled fixture. All rows are evaluated under the same downstream repair protocol; diagnosis-only front ends are explicitly paired with that shared repair loop when human repair success is reported. Self-consistency improves direct VLM reporting but increases cost. The validity column reports whether the emitted action stays inside the supported repair-plan vocabulary; it does not claim generator-adaptor executability. Best values are boldfaced; higher is better except for relative cost, where lower is better.

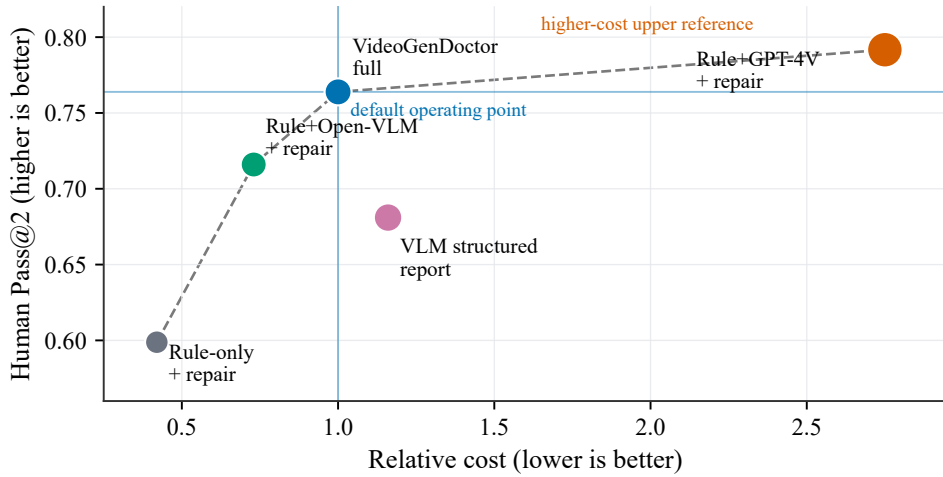
| Method                            | Macro-F1      | tIoU@0.5      | Human Pass@2  | Schema validity | Patch vocabulary validity | Relative cost |
|-----------------------------------|---------------|---------------|---------------|-----------------|---------------------------|---------------|
| VLM free-form report              | 0.8402        | 0.5986        | 0.6125        | 0.742           | 0.604                     | 1.12×         |
| VLM structured report             | 0.8869        | 0.6639        | 0.6810        | 0.914           | 0.732                     | 1.16×         |
| VLM structured + self-consistency | 0.8978        | 0.6824        | 0.7042        | 0.951           | 0.755                     | 3.48×         |
| VLM temporal proposal + patch     | 0.8915        | 0.7018        | 0.7076        | 0.936           | 0.802                     | 2.05×         |
| VideoGenDoctor-full               | 0.9110        | 0.7316        | 0.7639        | 0.991           | 0.953                     | <b>1.00</b> × |
| Rule+GPT-4V + repair loop         | <b>0.9276</b> | <b>0.7688</b> | <b>0.7917</b> | <b>0.993</b>    | <b>0.957</b>              | 2.75×         |



**Figure 6:** Closed-loop repair comparison on the controlled fixture. Left: automatic verifier Pass@1/2 on all 324 controlled videos. Right: blind human Pass@1/2 on the stratified 144-video human-evaluation subset. Automatic pass rates may be coupled to the system verifier, while human pass rates provide an independent check of repair success.

**Table 14:** Cost-performance trade-off under a shared downstream repair protocol. Diagnosis-only front ends are paired with the same repair loop before reporting Human Pass@2. Cost is reported relative to the default VideoGenDoctor-full configuration. Values in the last two columns are multiplicative factors relative to VideoGenDoctor-full (1.00×). Best values are boldfaced; higher is better for task metrics, lower is better for cost and latency.

| Method                      | Macro-F1      | tIoU@0.5      | Human Pass@2  | Relative cost | Relative latency |
|-----------------------------|---------------|---------------|---------------|---------------|------------------|
| Rule-only + repair loop     | 0.8926        | 0.6714        | 0.5988        | <b>0.42×</b>  | <b>0.58×</b>     |
| Rule+Open-VLM + repair loop | 0.9047        | 0.7062        | 0.7160        | 0.73×         | 0.87×            |
| VideoGenDoctor-full         | 0.9110        | 0.7316        | 0.7639        | 1.00×         | 1.00×            |
| Rule+GPT-4V + repair loop   | <b>0.9276</b> | <b>0.7688</b> | <b>0.7917</b> | 2.75×         | 2.31×            |
| VLM structured report       | 0.8869        | 0.6639        | 0.6810        | 1.16×         | 1.20×            |



**Figure 7:** Pareto view of the cost-performance trade-off under the shared downstream repair protocol. The default VideoGenDoctor-full operating point improves human repair success over lower-cost rule variants and direct structured VLM reporting, while the higher-cost Rule+GPT-4V reference obtains the strongest Human Pass@2 at substantially higher relative cost.

## B Additional Experimental Tables

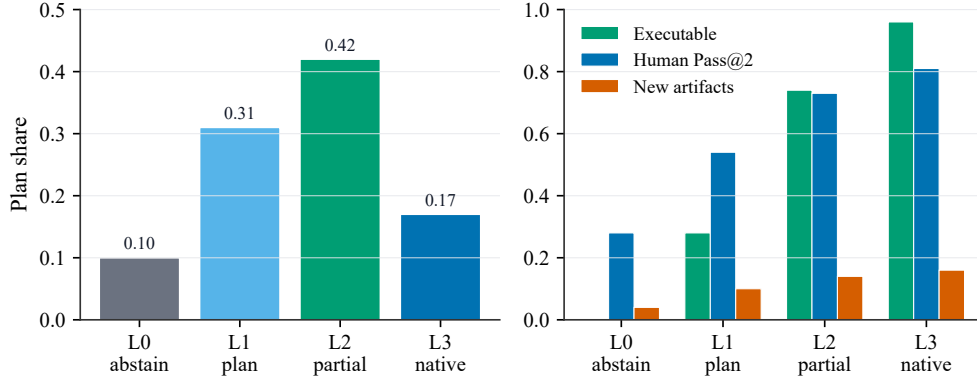
This appendix collects construction details, audit tables, stress analyses, and per-slice results that support the main experimental claims without interrupting the main narrative.

### B.1 Draft Extensions for Experiment Design

**Adapter executability.** Table 15 and Figure 8 separate repair-plan quality from backend executability by grouping generated plans into L0–L3 adapter levels. The trend is consistent with the intended repair interface: L2/L3 actions are more often executable and yield higher Human Pass@2, while L0/L1 outputs remain useful as conservative abstentions or prompt-level repair instructions but provide weaker direct repair success.

**Table 15:** Draft adapter-executability stratification. L0 denotes no-op or abstention; L1 denotes prompt-level repair instructions; L2 denotes partially executable repair actions; L3 denotes native generator/editor controls.

| Adapter level | Plan share | Adapter-executable rate | Human Pass@2 | New artifacts |
|---------------|------------|-------------------------|--------------|---------------|
| L0 abstain    | 0.100      | 0.000                   | 0.280        | 0.040         |
| L1 plan       | 0.310      | 0.280                   | 0.540        | 0.100         |
| L2 partial    | 0.420      | 0.740                   | 0.730        | 0.140         |
| L3 native     | 0.170      | 0.960                   | 0.810        | 0.160         |

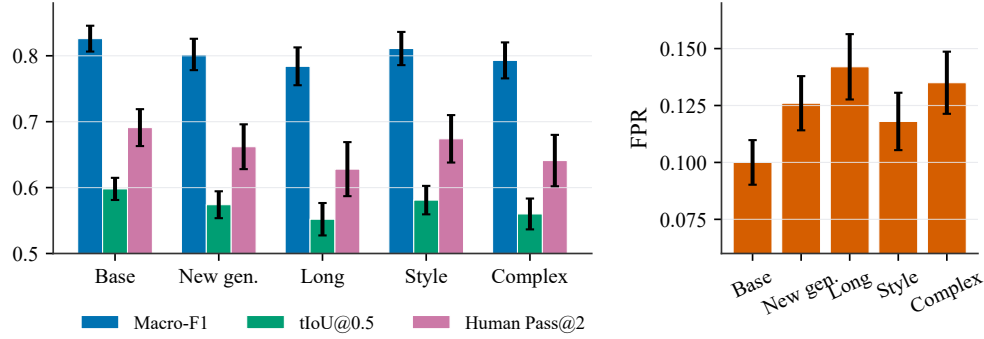


**Figure 8:** Adapter-executability stratification by repair-plan level. The left panel reports the share of generated plans by adapter level; the right panel compares adapter-executable rate, Human Pass@2, and new-artifact rate within each level.

**Stronger real-failure stress slices.** Table 16 and Figure 9 extend the real-failure check to harder slices: additional generators, longer videos, style-shifted prompts, and more complex prompt specifications. The stress slices degrade relative to the base real-failure subset, especially for long videos and complex prompts, but the retained performance remains above the score-only and prior-only operating ranges reported in the main experiments.

**Table 16:** Draft stress validation on harder real-failure slices. Intervals denote bootstrap-style half-widths used for plotting uncertainty in Figure 9.

| Slice             | Macro-F1          | tIoU@0.5          | Human Pass@2      | FPR               |
|-------------------|-------------------|-------------------|-------------------|-------------------|
| Base real-failure | $0.826 \pm 0.020$ | $0.598 \pm 0.017$ | $0.691 \pm 0.028$ | $0.100 \pm 0.010$ |
| New generators    | $0.802 \pm 0.024$ | $0.574 \pm 0.020$ | $0.662 \pm 0.034$ | $0.126 \pm 0.012$ |
| Long videos       | $0.784 \pm 0.029$ | $0.552 \pm 0.025$ | $0.628 \pm 0.041$ | $0.142 \pm 0.014$ |
| Style shift       | $0.811 \pm 0.025$ | $0.581 \pm 0.022$ | $0.674 \pm 0.036$ | $0.118 \pm 0.013$ |
| Complex prompts   | $0.793 \pm 0.027$ | $0.560 \pm 0.023$ | $0.641 \pm 0.039$ | $0.135 \pm 0.014$ |

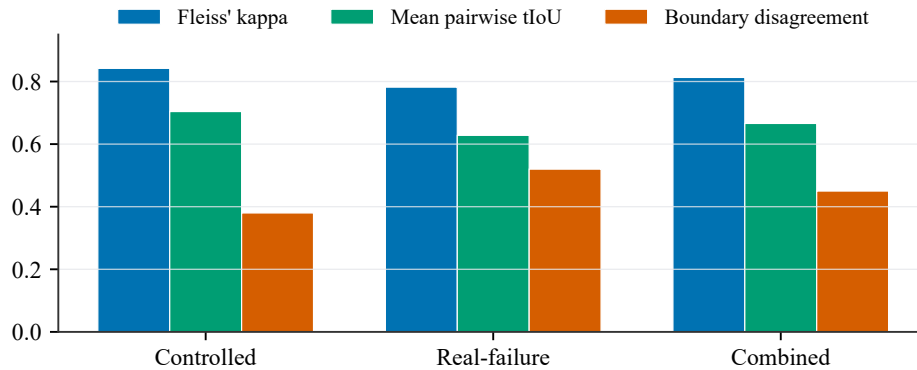


**Figure 9:** Draft stress validation on harder real-failure slices. Left: diagnosis, localization, and human repair metrics with uncertainty bars. Right: false-positive rate on matched real-normal controls for each slice.

331 **Multi-annotator stability.** Table 17 and Figure 10 extend the annotation reliability analysis from  
 332 dual annotation to a three-annotator draft setting. Code-level agreement remains substantial, while  
 333 span agreement is lower on real-failure samples, consistent with the higher ambiguity of naturally  
 334 occurring failures and less isolated temporal evidence.

**Table 17:** Draft three-annotator stability analysis. Boundary disagreement is measured in seconds; lower is better.

| Subset       | #Videos | #Candidate spans | Annotators | Fleiss' $\kappa$ | Mean pairwise tIoU / Boundary disagreement |
|--------------|---------|------------------|------------|------------------|--------------------------------------------|
| Controlled   | 72      | 468              | 3          | 0.842            | 0.704 / 0.38s                              |
| Real-failure | 48      | 300              | 3          | 0.782            | 0.628 / 0.52s                              |
| Combined     | 120     | 768              | 3          | 0.813            | 0.666 / 0.45s                              |



**Figure 10:** Draft multi-annotator stability analysis. Code-level Fleiss'  $\kappa$  is higher than temporal-boundary agreement, and the real-failure subset remains the most ambiguous split.



**Table 18:** Operator-template alignment analysis for the controlled fixture. The table makes explicit which observable cues, rule signals, failure codes, and repair-plan templates are aligned by construction.

| Perturbation operator                  | Observable cue                                             | Rule / feature signal                                  | Failure code                 | Repair-plan template                             | Alignment type      |
|----------------------------------------|------------------------------------------------------------|--------------------------------------------------------|------------------------------|--------------------------------------------------|---------------------|
| Identity-anchor removal                | face or body appearance drift                              | face embedding drift; character-region embedding drift | ID_FACE_DRIFT, ID_BODY_DRIFT | inject_reference_frame; replace_character_anchor | identity-anchor     |
| Required-prop dropping                 | required object becomes absent or inconsistent             | object detector absence; VLM prop-absence judgment     | AL_PROP_MISSING              | inject_prop                                      | prop-presence       |
| Camera-movement alteration             | requested trajectory is replaced by static or wrong motion | global flow direction; trajectory mismatch             | CA_MOVE_WRONG                | set_camera_movement                              | motion-control      |
| Shot-type / framing alteration         | framing violates requested shot type                       | crop statistics; VLM framing judgment                  | CA_SHOT_TYPE_WRONG           | adjust_framing                                   | framing-control     |
| Camera-shake injection                 | unstable camera with unintended jitter                     | flow instability; camera-shake heuristic               | CA_SHAKE                     | stabilize_camera                                 | stability-control   |
| Action dropping / shifting             | required event missing or misordered                       | action-track gap; VLM event-order judgment             | MO_EVENT_MISSING             | inject_action_keyframe; regenerate_segment       | event-structure     |
| Temporal jitter / frame-drop injection | discontinuity, duplicate frames, or rate mismatch          | flow anomaly; near-zero motion plateau                 | MO_JITTER, MO_SEGMENT_BREAK  | normalize_frame_rate; regenerate_segment         | temporal-continuity |
| Compression re-encoding                | blockiness and de-blocking artifacts                       | compression detector; texture degradation signal       | ST_COMPRESSION_ARTIFACT      | apply_enhancement                                | quality-restoration |
| Background-anchor weakening            | scene background drifts across the shot                    | scene embedding drift; background-anchor mismatch      | SC_BG_INCONSISTENCY          | fix_background_anchor                            | scene-anchor        |

**Table 19:** Source composition of the failure-enriched real-generator subset. We generate 480 raw videos and retain 240 samples with annotator-verified visible failures. Sampling is balanced at 60 retained failures per generator, and generator-specific prompt or conditioning adapters are logged in the construction manifest. The subset is used only as an initial transfer check, not to estimate natural failure prevalence. We use publicly identifiable generator checkpoints to make the subset construction auditable and reproducible.

| Generator              | Version / checkpoint | Input type                       | #Prompts / specs | #Videos | Avg. duration | #Unique observed codes |
|------------------------|----------------------|----------------------------------|------------------|---------|---------------|------------------------|
| CogVideoX              | CogVideoX1.5-5B      | text / ShotIR-to-prompt          | 60               | 60      | 8.4s          | 15                     |
| Wan                    | Wan2.1-T2V-14B       | text / ShotIR-to-prompt          | 60               | 60      | 9.2s          | 16                     |
| Stable Video Diffusion | SVD-XT-1.1           | image-to-video / reference frame | 60               | 60      | 8.6s          | 14                     |
| HunyuanVideo           | HunyuanVideo-1.5     | text / multi-shot prompt         | 60               | 60      | 9.5s          | 17                     |
| Total                  | –                    | –                                | 240              | 240     | 8.93s         | 18                     |

**Table 20:** Real-failure and real-normal evaluation. The real-normal subset contains 120 videos retained from the same 480 raw generations but judged to contain no obvious failure. Whenever a diagnosis-oriented front end is listed, patch-related columns are computed after pairing that front end with the shared downstream repair loop. Best values are boldfaced; higher is better except for real-normal FPR and unnecessary patch rate, where lower is better.

| Method                            | Real-failure Recall | Real-normal FPR | Unnecessary Patch Rate | No-op Accuracy | Specificity  |
|-----------------------------------|---------------------|-----------------|------------------------|----------------|--------------|
| Score-only                        | 0.621               | 0.258           | 0.231                  | 0.769          | 0.742        |
| Rule-only                         | 0.806               | 0.168           | 0.154                  | 0.846          | 0.832        |
| VLM structured report             | 0.818               | 0.217           | 0.205                  | 0.795          | 0.783        |
| VLM structured + self-consistency | 0.837               | 0.167           | 0.150                  | 0.850          | 0.833        |
| VLM temporal proposal + patch     | 0.848               | 0.192           | 0.183                  | 0.817          | 0.808        |
| VideoGenDoctor-diagnosis          | <b>0.862</b>        | 0.108           | 0.092                  | 0.908          | 0.892        |
| VideoGenDoctor-full               | 0.858               | <b>0.100</b>    | <b>0.087</b>           | <b>0.913</b>   | <b>0.900</b> |

**Table 21:** Stress analysis on 72 ambiguous specification-violation cases. These samples contain possible but non-adjudicable deviations from the specification. Whenever a diagnosis-oriented front end is listed, patch-related columns are computed after pairing that front end with the shared downstream repair loop. Best values are boldfaced; higher is better for abstention-oriented columns, lower is better for false concrete patch, wrong-target patch, over-repair, and new artifacts.

| Method                        | Abstention rate | Appropriate abstention | False concrete patch | Wrong-target patch | Over-repair  | New artifacts |
|-------------------------------|-----------------|------------------------|----------------------|--------------------|--------------|---------------|
| Score-only                    | 0.292           | 0.361                  | 0.514                | 0.222              | 0.347        | 0.083         |
| VLM structured report         | 0.181           | 0.292                  | 0.611                | 0.278              | 0.431        | 0.153         |
| VLM temporal proposal + patch | 0.222           | 0.347                  | 0.556                | 0.236              | 0.389        | 0.194         |
| VideoGenDoctor-diagnosis      | <b>0.500</b>    | <b>0.639</b>           | 0.292                | 0.125              | 0.208        | <b>0.069</b>  |
| VideoGenDoctor-full           | 0.472           | 0.625                  | <b>0.250</b>         | <b>0.111</b>       | <b>0.181</b> | 0.111         |

**Table 22:** Stress analysis on 48 low-quality or non-adjudicable cases. These samples are dominated by global degradation, unclear content, or insufficient evidence for reliable code-level labels. Whenever a diagnosis-oriented front end is listed, patch-related columns are computed after pairing that front end with the shared downstream repair loop. Best values are boldfaced; higher is better for quality-gated abstention, lower is better for the remaining columns.

| Method                        | Quality-gated abstention | False concrete patch | Wrong-target patch | Over-repair  | New artifacts |
|-------------------------------|--------------------------|----------------------|--------------------|--------------|---------------|
| Score-only                    | 0.417                    | 0.375                | 0.125              | 0.208        | 0.063         |
| VLM structured report         | 0.292                    | 0.458                | 0.188              | 0.271        | 0.104         |
| VLM temporal proposal + patch | 0.333                    | 0.396                | 0.167              | 0.250        | 0.146         |
| VideoGenDoctor-diagnosis      | <b>0.667</b>             | 0.146                | 0.063              | 0.104        | <b>0.042</b>  |
| VideoGenDoctor-full           | 0.625                    | <b>0.125</b>         | <b>0.042</b>       | <b>0.083</b> | 0.083         |

**Table 23:** Observed failure-code distribution in the controlled fixture and real-failure subset. The full taxonomy contains 38 codes, but empirical claims are restricted to the observed codes below.

| Failure code            | Group     | Controlled spans | Real-failure spans | Total spans |
|-------------------------|-----------|------------------|--------------------|-------------|
| ID_FACE_DRIFT           | Identity  | 150              | 112                | 262         |
| ID_BODY_DRIFT           | Identity  | 132              | 76                 | 208         |
| ID_CLOTHING_CHANGE      | Identity  | –                | 60                 | 60          |
| CA_MOVE_WRONG           | Camera    | 240              | 104                | 344         |
| CA_SHOT_TYPE_WRONG      | Camera    | 144              | 84                 | 228         |
| CA_SHAKE                | Camera    | 126              | 70                 | 196         |
| CA_WRONG_ANGLE          | Camera    | –                | 68                 | 68          |
| MO_JITTER               | Motion    | 174              | 124                | 298         |
| MO_FROZEN_FRAME         | Motion    | 156              | 72                 | 228         |
| MO_SEGMENT_BREAK        | Motion    | 162              | 88                 | 250         |
| MO_EVENT_MISSING        | Motion    | 126              | 90                 | 216         |
| MO_ACTION_ORDER_WRONG   | Motion    | –                | 56                 | 56          |
| AL_PROP_MISSING         | Alignment | 198              | 116                | 314         |
| AL_TEXT_PROMPT_MISMATCH | Alignment | –                | 132                | 132         |
| ST_COMPRESSION_ARTIFACT | Style     | 252              | 80                 | 332         |
| ST_COLOR_SHIFT          | Style     | –                | 56                 | 56          |
| SC_BG_INCONSISTENCY     | Scene     | 156              | 86                 | 242         |
| SC_OBJECT_TELEPORT      | Scene     | –                | 62                 | 62          |
| Total                   | –         | 2016             | 1536               | 3552        |

**Table 24:** Annotation reliability. Dual annotation is performed on a reliability subset before adjudication.

| Subset                          | #Videos | #Candidate spans | #Annotators | Code agreement   | Span agreement |
|---------------------------------|---------|------------------|-------------|------------------|----------------|
| Controlled reliability subset   | 72      | 468              | 2           | $\kappa = 0.858$ | tIoU = 0.716   |
| Real-failure reliability subset | 48      | 300              | 2           | $\kappa = 0.803$ | tIoU = 0.641   |
| Combined                        | 120     | 768              | 2           | $\kappa = 0.831$ | tIoU = 0.681   |

**Table 25:** Implementation details for VLM-based judges and external diagnosis/repair baselines.

| Setting                           | Model                  | Frames                    | Res.             | Prompt               | Rel. cost    |
|-----------------------------------|------------------------|---------------------------|------------------|----------------------|--------------|
| Rule+Open-VLM                     | Qwen2-VL-7B-Instruct   | 18                        | 512px short side | structured QA        | $0.34\times$ |
| Rule+GPT-4V                       | GPT-4V endpoint        | 18                        | 512px short side | structured QA        | $2.75\times$ |
| VLM free-form report              | GPT-4V endpoint        | 12                        | 512px short side | free-form critique   | $1.12\times$ |
| VLM structured report             | GPT-4V endpoint        | 12                        | 512px short side | JSON schema          | $1.16\times$ |
| VLM structured + self-consistency | GPT-4V endpoint        | 12 frames $\times$ 3 runs | 512px short side | JSON schema, 3 runs  | $3.48\times$ |
| VLM temporal proposal + patch     | GPT-4V endpoint        | 12                        | 512px short side | span proposal + JSON | $2.05\times$ |
| VLM-to-patch                      | GPT-4V endpoint        | 12                        | 512px short side | JSON patch           | $1.19\times$ |
| Score+LLM-to-patch                | scalar evaluator + LLM | 0                         | –                | score-to-patch JSON  | N/A          |

*Note.* Most direct VLM baselines share the same hosted GPT-4V endpoint OpenAI [2023] to isolate prompt- and schema-level effects rather than backbone changes; Rule+Open-VLM provides an open-model reference. Hosted endpoint calls are logged with raw responses for auditability. ‘N/A’ denotes a scalar-evaluator-dominated pipeline whose cost is not directly comparable to frame-based VLM calls.

**Table 26:** Per-code reliability of the default VideoGenDoctor diagnosis configuration on the controlled fixture.

| Observed code           | Precision | Recall | F1    | tIoU@0.5 |
|-------------------------|-----------|--------|-------|----------|
| ID_FACE_DRIFT           | 0.938     | 0.914  | 0.926 | 0.748    |
| ID_BODY_DRIFT           | 0.922     | 0.897  | 0.909 | 0.725    |
| CA_MOVE_WRONG           | 0.932     | 0.916  | 0.924 | 0.734    |
| CA_SHOT_TYPE_WRONG      | 0.913     | 0.889  | 0.901 | 0.705    |
| CA_SHAKE                | 0.901     | 0.878  | 0.889 | 0.692    |
| MO_JITTER               | 0.890     | 0.865  | 0.877 | 0.676    |
| MO_FROZEN_FRAME         | 0.963     | 0.950  | 0.956 | 0.794    |
| MO_SEGMENT_BREAK        | 0.906     | 0.883  | 0.894 | 0.704    |
| MO_EVENT_MISSING        | 0.875     | 0.848  | 0.861 | 0.662    |
| AL_PROP_MISSING         | 0.951     | 0.930  | 0.940 | 0.760    |
| ST_COMPRESSION_ARTIFACT | 0.950     | 0.966  | 0.958 | 0.779    |
| SC_BG_INCONSISTENCY     | 0.910     | 0.885  | 0.897 | 0.686    |

**Table 27:** Per-code reliability on the real-failure subset. The subset is more challenging than the controlled fixture, especially for motion, camera, and scene-level failures.

| Failure code            | #Spans | Precision | Recall | F1    | tIoU@0.5 |
|-------------------------|--------|-----------|--------|-------|----------|
| ID_FACE_DRIFT           | 112    | 0.884     | 0.848  | 0.866 | 0.648    |
| ID_BODY_DRIFT           | 76     | 0.856     | 0.823  | 0.839 | 0.615    |
| ID_CLOTHING_CHANGE      | 60     | 0.834     | 0.798  | 0.816 | 0.586    |
| CA_MOVE_WRONG           | 104    | 0.866     | 0.832  | 0.849 | 0.628    |
| CA_SHOT_TYPE_WRONG      | 84     | 0.838     | 0.803  | 0.820 | 0.590    |
| CA_SHAKE                | 70     | 0.820     | 0.779  | 0.799 | 0.559    |
| CA_WRONG_ANGLE          | 68     | 0.825     | 0.786  | 0.805 | 0.568    |
| MO_JITTER               | 124    | 0.823     | 0.777  | 0.799 | 0.565    |
| MO_FROZEN_FRAME         | 72     | 0.899     | 0.854  | 0.876 | 0.663    |
| MO_SEGMENT_BREAK        | 88     | 0.827     | 0.792  | 0.809 | 0.582    |
| MO_EVENT_MISSING        | 90     | 0.807     | 0.762  | 0.784 | 0.542    |
| MO_ACTION_ORDER_WRONG   | 56     | 0.792     | 0.748  | 0.769 | 0.524    |
| AL_PROP_MISSING         | 116    | 0.893     | 0.863  | 0.878 | 0.669    |
| AL_TEXT_PROMPT_MISMATCH | 132    | 0.842     | 0.794  | 0.817 | 0.596    |
| ST_COMPRESSION_ARTIFACT | 80     | 0.914     | 0.874  | 0.893 | 0.688    |
| ST_COLOR_SHIFT          | 56     | 0.846     | 0.806  | 0.825 | 0.601    |
| SC_BG_INCONSISTENCY     | 86     | 0.832     | 0.795  | 0.813 | 0.578    |
| SC_OBJECT_TELEPORT      | 62     | 0.823     | 0.782  | 0.802 | 0.545    |

**Table 28:** Per-generator results on the real-failure and real-normal subsets. FPR is computed on 30 real-normal samples per generator.

| Generator              | Macro-F1 | tIoU@0.5 | Human Pass@2 | Real-normal FPR | Main failure groups       | Normal samples |
|------------------------|----------|----------|--------------|-----------------|---------------------------|----------------|
| CogVideoX              | 0.834    | 0.609    | 0.702        | 0.092           | Camera, Motion, Alignment | 30             |
| Wan                    | 0.821    | 0.588    | 0.681        | 0.108           | Motion, Camera, Identity  | 30             |
| Stable Video Diffusion | 0.842    | 0.617    | 0.696        | 0.083           | Identity, Style, Scene    | 30             |
| HunyuanVideo           | 0.809    | 0.579    | 0.684        | 0.117           | Camera, Motion, Alignment | 30             |
| Average                | 0.827    | 0.598    | 0.691        | 0.100           | –                         | 120            |

**Table 29:** Leave-one-perturbation-out evaluation. Each held-out operator is excluded from threshold tuning and then used only for testing.

| Held-out perturbation operator         | Macro-F1 | tIoU@0.5 | Top-3 hit |
|----------------------------------------|----------|----------|-----------|
| Identity-anchor removal                | 0.884    | 0.681    | 0.644     |
| Required-prop dropping                 | 0.902    | 0.704    | 0.672     |
| Camera-movement alteration             | 0.871    | 0.653    | 0.612     |
| Shot-type / framing alteration         | 0.862    | 0.641    | 0.606     |
| Camera-shake injection                 | 0.851    | 0.628    | 0.590     |
| Action dropping / shifting             | 0.838    | 0.612    | 0.568     |
| Temporal jitter / frame-drop injection | 0.846    | 0.621    | 0.577     |
| Compression re-encoding                | 0.910    | 0.719    | 0.688     |
| Background-anchor weakening            | 0.868    | 0.646    | 0.610     |
| Average                                | 0.870    | 0.656    | 0.619     |

**Table 30:** Leave-one-source-group-out evaluation over the six controlled-fixture domain groups.

| Held-out source group      | Macro-F1 | tIoU@0.5 | Top-3 hit |
|----------------------------|----------|----------|-----------|
| Indoor / human-object      | 0.899    | 0.714    | 0.665     |
| Outdoor / motion           | 0.887    | 0.702    | 0.651     |
| Multi-object interaction   | 0.892    | 0.706    | 0.657     |
| Camera-control scene       | 0.875    | 0.683    | 0.631     |
| Style / lighting variation | 0.907    | 0.722    | 0.676     |
| Scene-change / background  | 0.884    | 0.694    | 0.642     |
| Average                    | 0.891    | 0.704    | 0.654     |

## 335 C Full Failure Taxonomy

336 Table 31 reports the machine-readable taxonomy used by VideoGenDoctor v0.1. The v0.1 taxonomy  
337 contains 38 codes grouped into six categories. Each code is paired with an operational definition,  
338 an evidence procedure, and a default patch template. The current benchmark evaluates the 12 codes  
339 observed in VideoGenDoctor-Bench-v0, while the remaining codes define the extensible namespace  
340 used by the report schema and patch compiler.

**Table 31:** Full VideoGenDoctor v0.1 failure taxonomy used by the report schema and patch compiler.

| Code               | Definition                                                       | Evidence Procedure                                                       | Patch Template                   |
|--------------------|------------------------------------------------------------------|--------------------------------------------------------------------------|----------------------------------|
| Identity           |                                                                  |                                                                          |                                  |
| ID_FACE_DRIFT      | Face identity changes inconsistently between frames or segments. | Compare face embeddings across keyframes and flag large cosine distance. | inject_<br>reference_<br>frame   |
| ID_BODY_DRIFT      | Body appearance, clothing, or build changes inconsistently.      | Compare character-region embeddings across segments.                     | inject_<br>reference_<br>frame   |
| ID_WRONG_CHARACTER | A visible character does not match any specified character.      | Check face embeddings against known character anchors.                   | replace_<br>character_<br>anchor |

Continued on next page

| Code                    | Definition                                                              | Evidence Procedure                                                        | Patch Template            |
|-------------------------|-------------------------------------------------------------------------|---------------------------------------------------------------------------|---------------------------|
| ID_CLOTHING_CHANGE      | Clothing changes between shots without narrative justification.         | Measure character-region appearance drift across segments.                | fix_character_consistency |
| ID_MULTIPLE_FACES       | Multiple distinct faces appear when one character is expected.          | Count distinct identities detected in a single-character shot.            | adjust_character_count    |
| ID_NO_FACE_VISIBLE      | An expected face is absent when the shot requires face visibility.      | Check face detection under close-up or medium-shot constraints.           | adjust_framing            |
| Scene                   |                                                                         |                                                                           |                           |
| SC_BG_INCONSISTENCY     | The background changes unexpectedly between segments.                   | Measure background-region embedding drift across consecutive segments.    | fix_background_anchor     |
| SC_ENVIRONMENT_MISMATCH | The scene environment does not match the specified location.            | Compare frames with the location description using image-text similarity. | set_environment           |
| SC_LIGHTING_SHIFT       | Lighting changes inconsistently within a shot.                          | Measure per-frame luminance variation within the segment.                 | normalize_lighting        |
| SC_OBJECT_TELEPORT      | A scene object appears or disappears without plausible motion.          | Track object boxes across adjacent frames.                                | stabilize_scene_objects   |
| SC_BG_FLICKER           | The background flickers or pulses unnaturally.                          | Detect high-frequency luminance variation in background regions.          | fix_background_anchor     |
| SC_WRONG_TIME_OF_DAY    | The time of day does not match the specification.                       | Classify sky and ambient lighting against the expected time.              | set_time_of_day           |
| Motion                  |                                                                         |                                                                           |                           |
| MO_JITTER               | Unnatural frame-to-frame jitter appears outside intended camera motion. | Measure optical-flow magnitude variance and direction instability.        | normalize_frame_rate      |
| MO_FRAME_DROP           | Duplicate or dropped frames produce visible stutter.                    | Detect near-zero adjacent-frame flow when motion is expected.             | interpolate_frames        |
| MO_UNNATURAL_MOTION     | Character or object motion is physically implausible.                   | Compare flow direction and magnitude with expected motion constraints.    | constrain_motion          |
| MO_EVENT_MISSING        | A required action or event does not occur in the expected segment.      | Ask the VLM judge whether the specified action is visible.                | inject_action_keyframe    |
| MO_ACTION_ORDER_WRONG   | Actions occur in the wrong temporal order.                              | Compare detected action timestamps with the specification order.          | reorder_segments          |
| MO_MOTION_BLUR_EXCESS   | Excessive motion blur degrades keyframes.                               | Use sharpness measures such as Laplacian variance.                        | reduce_motion_blur        |
| MO_FROZEN_FRAME         | The video is frozen when motion is expected.                            | Detect near-zero flow across the full segment.                            | regenerate_segment        |
| MO_SEGMENT_BREAK        | A visual discontinuity occurs at a segment boundary.                    | Detect embedding-drift spikes at boundaries.                              | smooth_segment_transition |
| Camera                  |                                                                         |                                                                           |                           |
| CA_MOVE_WRONG           | Camera movement deviates from the specified movement.                   | Compare optical-flow direction patterns with the expected movement type.  | set_camera_movement       |
| CA_SHOT_TYPE_WRONG      | The shot type does not match the specification.                         | Compare face or body box size ratios with the expected shot type.         | adjust_framing            |
| CA_UNINTENDED_ZOOM      | Unplanned zoom-in or zoom-out occurs.                                   | Detect radial flow without a zoom specification.                          | remove_zoom               |
| CA_SHAKE                | Camera shake appears outside the specified movement.                    | Detect high-frequency oscillation in global optical flow.                 | stabilize_camera          |
| CA_WRONG_ANGLE          | Camera angle deviates from the specification.                           | Ask the VLM judge to compare the observed angle with the target angle.    | set_camera_angle          |
| CA_ROLL                 | Unintended camera roll appears.                                         | Estimate the rotational component of optical flow.                        | remove_camera_roll        |
| Alignment               |                                                                         |                                                                           |                           |
| AL_PROP_MISSING         | A required prop is not visible in the segment.                          | Use object detection or VLM verification for prop absence.                | inject_prop               |
| AL_PROP_WRONG_PLACEMENT | A required prop appears in the wrong region.                            | Compare the prop box with the expected screen region.                     | reposition_prop           |

Continued on next page

| Code                       | Definition                                                         | Evidence Procedure                                                | Patch Template             |
|----------------------------|--------------------------------------------------------------------|-------------------------------------------------------------------|----------------------------|
| AL_TEXT_PROMPT_MISMATCH    | The generated content does not align with the prompt.              | Measure text-image similarity and verify mismatches with a judge. | strengthen_prompt_guidance |
| AL_CHARACTER_COUNT_WRONG   | The number of visible characters does not match the specification. | Count faces or bodies and compare with the expected count.        | adjust_character_count     |
| AL_WRONG_PROP              | A visible prop is not listed in the specification.                 | Detect high-confidence unlisted objects.                          | remove_prop                |
| AL_PROP_OCCLUDED           | A required prop is present but substantially occluded.             | Estimate visible area and occlusion of the prop box.              | reposition_prop            |
| Style                      |                                                                    |                                                                   |                            |
| ST_COLOR_SHIFT             | The color palette changes inconsistently across segments.          | Measure HSV histogram distance between segments.                  | apply_color_grading        |
| ST_ART_STYLE_INCONSISTENCY | The art style shifts between segments.                             | Compare style embeddings across segments.                         | apply_style_transfer       |
| ST_COMPRESSION_ARTIFACT    | Visible compression artifacts degrade video quality.               | Detect blocking, ringing, or quality-score degradation.           | apply_enhancement          |
| ST_TEXTURE_FLICKER         | Texture patterns flicker or shimmer unnaturally.                   | Measure high temporal frequency in texture regions.               | smooth_texture             |
| ST_OVEREXPOSURE            | Significant frame regions are overexposed.                         | Count the fraction of saturated high-value pixels.                | adjust_exposure            |
| ST_UNDEREXPOSURE           | Significant frame regions are underexposed.                        | Count the fraction of near-black pixels.                          | adjust_exposure            |

## D Judge Protocol

The Stage-2 judge receives the top- $K$  candidate segments produced by Stage-1. For each segment, the input includes the segment identifier, temporal bounds, keyframe paths, candidate failure codes, and specification context, including required props, expected actions, camera movement, shot type, and character anchors. The judge answers structured questions rather than producing a free-form report. This design keeps the output comparable across local open-source VLMs and hosted vision-capable models.

For identity failures, the judge checks whether the face, body appearance, clothing, or other distinctive visual attributes of the main character change inconsistently across keyframes. For alignment failures, it verifies whether required props are visible, whether they appear in the expected screen region, and whether the generated content matches the prompt or structured specification. For camera failures, it compares the observed camera movement, shot type, and viewpoint angle with the requested control signal. For motion failures, it verifies whether required actions occur, whether they occur in the specified order, and whether stutter, frozen frames, or segment breaks are visible. For style and scene failures, it checks lighting, color, compression artifacts, background consistency, and environment match.

The output is a structured object containing a natural-language answer for each question, a confidence score, per-code probability updates, and a re-ranking of keyframes. The final failure score is computed as

$$\tilde{\sigma}_{s,c} = (1 - \alpha)\sigma_{s,c}^{(1)} + \alpha\sigma_{s,c}^{(2)},$$

where  $\sigma_{s,c}^{(1)}$  is the Stage-1 score,  $\sigma_{s,c}^{(2)}$  is the judge score, and  $\alpha = 0.6$  by default. All judge calls record the model name, prompt, raw response, token count when available, and output path to support reproducibility.

## E Patch Template Examples

VideoGenDoctor compiles each diagnosis record into one or more patch actions. If a ShotIR specification is available, the compiler emits field-level diffs. If the generator does not expose a structured specification interface, the compiler emits generator-agnostic repair plans that can be translated into prompt edits, reference-frame injection, segment-level re-rendering, or post-processing steps.

```
{"action": "inject_reference_frame",
  "field": "character_id",
```

```

371 "value": "<anchor_frame>",
372 "reason": "ID_FACE_DRIFT detected at t=1.2s"}

373 {"action": "set_camera_movement",
374 "field": "camera_movement",
375 "value": "<expected_camera_movement>",
376 "reason": "CA_MOVE_WRONG detected in the localized span"}

377 {"action": "inject_prop",
378 "field": "props_required",
379 "value": "<missing_prop>",
380 "reason": "AL_PROP_MISSING verified by object or VLM evidence"}

381 {"action": "regenerate_segment",
382 "field": "segment",
383 "value": "<t0,t1>",
384 "reason": "MO_FROZEN_FRAME or MO_SEGMENT_BREAK is localized to this span"}

385 Each patch specifies an action type, the affected target, an optional update value, and a short rationale
386 tied to the evidence span. When multiple records affect the same target, the compiler groups them
387 before re-rendering to reduce conflicts among patch actions.

```

**Table 32:** Patch-to-generator adapter details. A repair plan is counted as adapter-executable only when it can be automatically translated into target-generator or editor inputs without manual rewriting. L0/L1 rows denote abstention or prompt-rewrite instructions rather than executable patches.

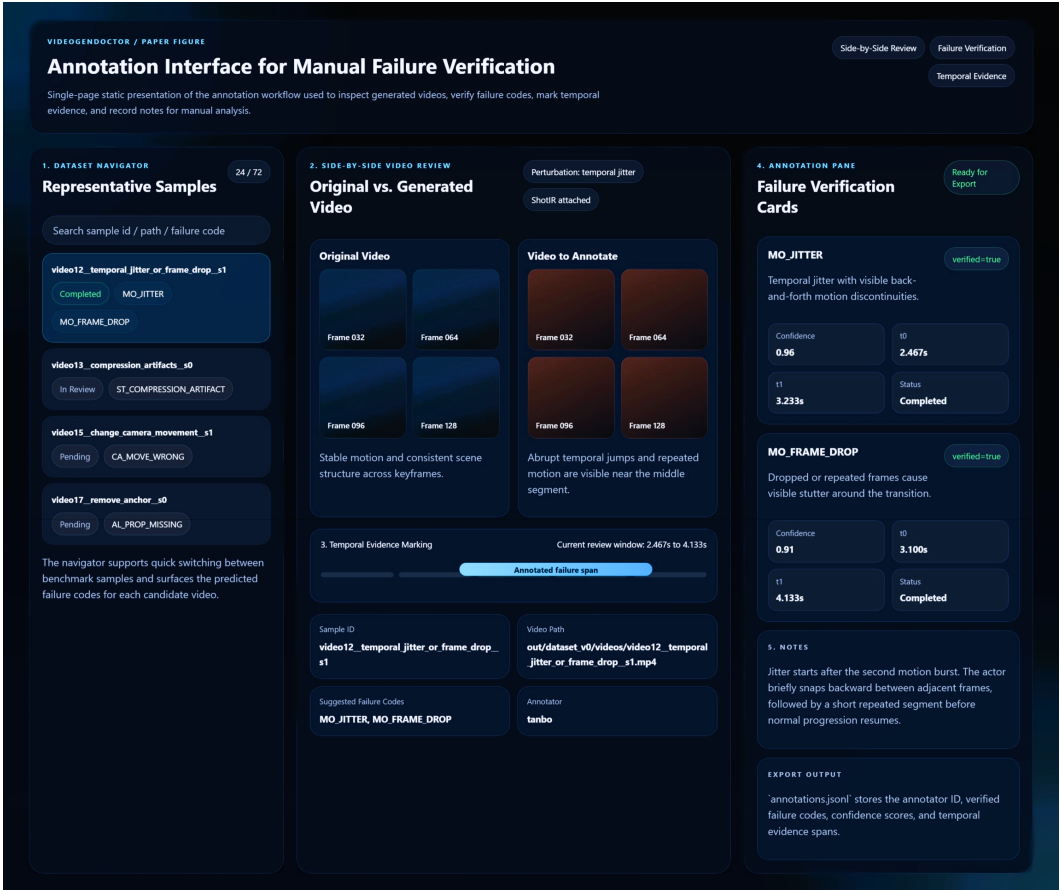
| Repair-plan action     | ShotIR field              | Counted as | Adapter availability                                      | Concrete generator/editor input                                  | Fallback                                    |
|------------------------|---------------------------|------------|-----------------------------------------------------------|------------------------------------------------------------------|---------------------------------------------|
| inject_reference_frame | character.identity_anchor | L2         | L2 available for SVD; L1 partial for text-to-video models | reference image input + identity prompt strengthening            | prompt-only repair plan                     |
| set_camera_movement    | camera.movement           | L1         | L1 partial                                                | camera-control prompt rewrite; trajectory adapter when supported | no_op_uncertain if no camera control exists |
| set_camera_angle       | camera.viewpoint          | L1         | L1 partial                                                | viewpoint prompt rewrite; optional camera-control token          | repair suggestion only                      |
| inject_prop            | props.required            | L1 / L2    | L1 available; L2 partial with mask/reference input        | prop phrase insertion + optional reference or mask               | prompt edit                                 |
| reposition_prop        | props.region              | L1         | L1 partial                                                | region-aware prompt rewrite or mask-conditioned edit             | prompt-only repair plan                     |
| regenerate_segment     | temporal span             | L2 / L3    | L2 available when segment rerendering is supported        | segment-level rerender or temporal inpainting call               | full-clip rerender                          |
| stabilize_camera       | camera.constraint         | L1 / L2    | L1 partial                                                | no-shake constraint or stabilization post-process                | repair suggestion only                      |
| normalize_frame_rate   | motion.continuity         | L2         | L2 partial                                                | frame interpolation or temporal smoothing post-process           | regenerate_segment                          |
| apply_enhancement      | style.quality             | L2         | L2 available                                              | enhancement / deblocking post-process                            | prompt edit for higher quality              |
| fix_background_anchor  | scene.background_anchor   | L1         | L1 partial                                                | background-anchor prompt strengthening; optional reference image | prompt-only repair plan                     |
| no_op_uncertain        | –                         | L0         | L0 available                                              | abstain from concrete repair                                     | no action                                   |

388 In the reported Human Pass@K evaluation, the concretely executed actions are the L2/L3 subset  
389 exposed by the active backend, primarily reference-frame injection, segment rerendering or temporal  
390 smoothing, enhancement or deblocking, and mask- or reference-conditioned prop insertion when  
391 supported. L0/L1 rows remain structured repair-plan instructions and are not counted as executable  
392 patches.

## 393 F Annotation Guide Summary

394 Annotators review each perturbed video together with the original prompt or ShotIR specification  
395 and the auto-assigned candidate failure codes. For each candidate, annotators determine whether  
396 the failure is visually present, assign the tightest temporal span that contains the visible deviation,

and select representative keyframes when available. Annotators may also add missing failures if the perturbation produces an additional visible artifact.



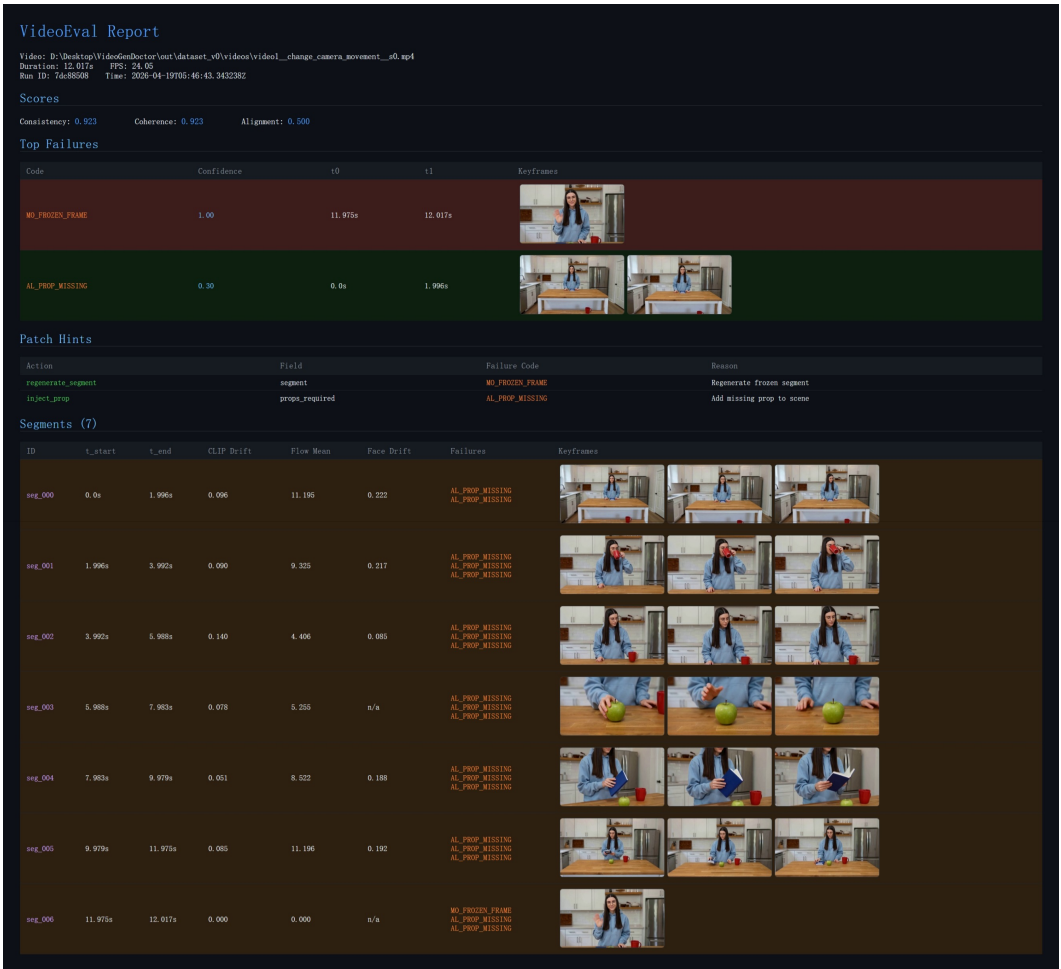
**Figure 11:** Annotation interface used for manual verification in VideoGenDoctor-Bench-v0. The tool exposes a sample navigator, side-by-side review of the original and perturbed videos, temporal evidence marking, structured failure verification cards, and free-form notes for adjudication.

The annotation record is stored as one JSON object per video and includes the video identifier, annotator identifier, verified failure codes, confidence values, evidence spans, and free-form notes. A failure is retained in the benchmark only when the annotator can verify it from the rendered video and specification, without relying on perturbation metadata alone. Ambiguous cases are handled conservatively: if a deviation is not visually identifiable, the candidate is marked as not verified.

The experiment fixture includes a dual-annotated reliability subset of 120 videos and 768 candidate evidence spans. The subset contains both controlled perturbation samples and naturally occurring real-failure samples. Larger multi-annotator validation with more than two independent annotators remains future work.



408 **G Example Diagnostic Report**



**Figure 12:** Example VideoGenDoctor diagnostic report for a perturbed video. The report summarizes global scores, ranked failure records with confidence and temporal spans, representative keyframes, patch hints derived from the retained failure codes, and segment-level evidence used for localized review and repair.

409 **H Direct VLM Baseline Prompt Templates**

410 The four direct VLM/LLM baselines differ only in their allowed information access and output schema.  
411 The free-form and structured VLM report baselines receive sampled frames and the specification.  
412 The VLM-to-patch baseline also receives sampled frames, the taxonomy names, and the repair-plan  
413 vocabulary, but not VideoGenDoctor’s rule scores or patch compiler outputs. The Score+LLM-to-  
414 patch baseline receives scalar evaluator scores and the specification only; it does not receive sampled  
415 frames. These information boundaries are fixed before evaluation and are used for all samples.

416 **F.1 Free-form VLM report prompt.** The model receives the video specification, sampled  
417 keyframes with timestamps, and a short instruction: identify visible failures, explain the evidence,  
418 and propose repair suggestions. The response is then parsed into taxonomy codes and evidence spans  
419 for evaluation.

420 **F.2 Structured VLM report prompt.** The model receives the same input but must output  
421 JSON objects with fields failure\_code, temporal\_span, affected\_target, confidence,  
422 evidence\_keyframes, rationale, and repair\_action. Failures without visual evidence must  
423 be marked as uncertain rather than hallucinated.

```

424 You are evaluating whether a generated video follows the given specification.
425 For each visible failure, return a JSON record:
426 {
427     "failure_code": "<taxonomy code>",
428     "temporal_span": [t0, t1],
429     "affected_target": "<character / prop / camera / motion / style / scene>",
430     "confidence": <0-1>,
431     "evidence_keyframes": ["frame@time", ...],
432     "rationale": "short visual reason",
433     "repair_action": "<patch action>"
434 }
435 Only report failures that are visually supported by the frames.

```

**F.3 VLM-to-patch prompt.** The VLM-to-patch baseline receives the same video specification and sampled keyframes as the direct VLM diagnosis baselines, but it is asked to output repair actions directly rather than diagnosis records. This baseline is allowed to access the taxonomy names and the supported repair-plan vocabulary, but it is not given VideoGenDoctor’s Stage-1 rule scores, candidate segments, or patch compiler outputs. It may infer temporal spans from the sampled frames, but it does not receive ground-truth spans or VideoGenDoctor-predicted spans. This setting tests whether a VLM can directly produce structured repair plans from the specification and visual evidence.

```

443 You are a video-generation repair planner.
444
445 Input:
446 1. A video specification or ShotIR-style instruction.
447 2. Sampled keyframes from the generated video, each with a timestamp.
448 3. The failure-code taxonomy names.
449 4. The supported repair-plan vocabulary.
450
451 Your task:
452 Inspect the sampled frames and the specification. Identify only visually supported
453 problems and output structured repair-plan actions. Do not write a free-form critique.
454 Do not invent failures that are not visible in the frames.
455
456 You may use the following failure-code groups:
457 - Identity: ID_FACE_DRIFT, ID_BODY_DRIFT, ID_WRONG_CHARACTER,
458   ID_CLOTHING_CHANGE, ID_MULTIPLE_FACES, ID_NO_FACE_VISIBLE
459 - Camera: CA_MOVE_WRONG, CA_SHOT_TYPE_WRONG, CA_UNINTENDED_ZOOM,
460   CA_SHAKE, CA_WRONG_ANGLE, CA_ROLL
461 - Motion: MO_JITTER, MO_FRAME_DROP, MO_UNNATURAL_MOTION,
462   MO_EVENT_MISSING, MO_ACTION_ORDER_WRONG, MO_MOTION_BLUR_EXCESS,
463   MO_FROZEN_FRAME, MO_SEGMENT_BREAK
464 - Alignment: AL_PROP_MISSING, AL_PROP_WRONG_PLACEMENT,
465   AL_TEXT_PROMPT_MISMATCH, AL_CHARACTER_COUNT_WRONG,
466   AL_WRONG_PROP, AL_PROP_OCCLUDED
467 - Style: ST_COLOR_SHIFT, ST_ART_STYLE_INCONSISTENCY,
468   ST_COMPRESSION_ARTIFACT, ST_TEXTURE_FLICKER,
469   ST_OVEREXPOSURE, ST_UNDEREXPOSURE
470 - Scene: SC_BG_INCONSISTENCY, SC_ENVIRONMENT_MISMATCH,
471   SC_LIGHTING_SHIFT, SC_OBJECT_TELEPORT, SC_BG_FLICKER,
472   SC_WRONG_TIME_OF_DAY
473
474 Supported repair-plan vocabulary:
475 - inject_reference_frame
476 - replace_character_anchor
477 - fix_character_consistency
478 - adjust_framing
479 - set_camera_movement
480 - set_camera_angle

```

```

481 - stabilize_camera
482 - remove_zoom
483 - remove_camera_roll
484 - inject_prop
485 - reposition_prop
486 - remove_prop
487 - adjust_character_count
488 - strengthen_prompt_guidance
489 - inject_action_keyframe
490 - reorder_segments
491 - regenerate_segment
492 - interpolate_frames
493 - normalize_frame_rate
494 - constrain_motion
495 - smooth_segment_transition
496 - reduce_motion_blur
497 - apply_color_grading
498 - apply_style_transfer
499 - apply_enhancement
500 - smooth_texture
501 - adjust_exposure
502 - fix_background_anchor
503 - set_environment
504 - normalize_lighting
505 - stabilize_scene_objects
506 - set_time_of_day
507 - no_op_uncertain
508
509 Return a JSON object with the following schema:
510 {
511     "patches": [
512         {
513             "patch_id": "<unique patch id>",
514             "failure_code": "<taxonomy code or uncertain>",
515             "action": "<one item from the supported repair-plan vocabulary>",
516             "target": "<character / prop / camera / motion / style / scene target>",
517             "temporal_span": [<start_sec>, <end_sec>],
518             "evidence_keyframes": ["frame@time", "..."],
519             "confidence": <0-1>,
520             "patch_fields": {
521                 "field": "<generator or ShotIR field to modify>",
522                 "value": "<new value, constraint, or reference to insert>"
523             },
524             "rationale": "<short visual reason>"
525         }
526     ],
527     "global_notes": "<one-sentence summary>",
528     "uncertain": <true or false>
529 }
530
531 Rules:
532 1. Output valid JSON only.
533 2. Every patch must use an action from the supported repair-plan vocabulary.
534 3. If no visually supported failure is visible, return:
535     {"patches": [], "global_notes": "No visually supported repair action.", "uncertain": true}
536 4. If the temporal span is uncertain, estimate it from the nearest sampled frame
537     timestamps and set "confidence" below 0.5.
538 5. Do not output patches for purely aesthetic preferences unless the specification
539     explicitly requires them.

```

540 **F4 Score+LLM-to-patch prompt.** The Score+LLM-to-patch baseline receives scalar evaluator  
 541 outputs and the original video specification, but it does not receive sampled frames. This baseline is  
 542 intentionally weaker in visual grounding but represents a common production setting where low-level  
 543 evaluators return scalar scores and an LLM converts low-scoring dimensions into repair suggestions.  
 544 Because it cannot inspect frames, it is not allowed to claim frame-specific visual evidence. It may  
 545 output approximate temporal spans only when the scalar evaluator provides segment-level scores;  
 546 otherwise the temporal span must be marked as null. This makes the information boundary explicit  
 547 and prevents unfair comparison with frame-based VLM baselines.

548 You are a repair planner for controllable video generation.

549

550 Input:

- 551 1. The original video specification or ShotIR-style instruction.
- 552 2. Scalar evaluator outputs, such as quality, alignment, motion, identity,  
 553 camera, style, and scene-consistency scores.
- 554 3. Optional segment-level scalar scores if available.
- 555 4. The supported repair-plan vocabulary.

556

557 Important information boundary:

558 You do NOT receive sampled frames or the video itself. Therefore:

- 559 - Do not claim that a failure is visually observed.
- 560 - Do not mention specific visual evidence or keyframes.
- 561 - Only infer likely repair actions from low-scoring evaluator dimensions.
- 562 - Use "evidence\_source": "scalar\_score" for all patches.
- 563 - Use "temporal\_span": null unless segment-level scores identify a specific interval.

564

565 Supported repair-plan vocabulary:

- 566 - inject\_reference\_frame
- 567 - replace\_character\_anchor
- 568 - fix\_character\_consistency
- 569 - adjust\_framing
- 570 - set\_camera\_movement
- 571 - set\_camera\_angle
- 572 - stabilize\_camera
- 573 - remove\_zoom
- 574 - remove\_camera\_roll
- 575 - inject\_prop
- 576 - reposition\_prop
- 577 - remove\_prop
- 578 - adjust\_character\_count
- 579 - strengthen\_prompt\_guidance
- 580 - inject\_action\_keyframe
- 581 - reorder\_segments
- 582 - regenerate\_segment
- 583 - interpolate\_frames
- 584 - normalize\_frame\_rate
- 585 - constrain\_motion
- 586 - smooth\_segment\_transition
- 587 - reduce\_motion\_blur
- 588 - apply\_color\_grading
- 589 - apply\_style\_transfer
- 590 - apply\_enhancement
- 591 - smooth\_texture
- 592 - adjust\_exposure
- 593 - fix\_background\_anchor
- 594 - set\_environment
- 595 - normalize\_lighting
- 596 - stabilize\_scene\_objects
- 597 - set\_time\_of\_day

```

598 - no_op_uncertain
599
600 Return a JSON object with the following schema:
601 {
602   "patches": [
603     {
604       "patch_id": "<unique patch id>",
605       "score_dimension": "<low-scoring evaluator dimension>",
606       "inferred_failure_type": "<short inferred failure description>",
607       "action": "<one item from the supported repair-plan vocabulary>",
608       "target": "<character / prop / camera / motion / style / scene target>",
609       "temporal_span": null,
610       "evidence_source": "scalar_score",
611       "triggering_scores": {
612         "<score_name>": <score_value>
613       },
614       "confidence": <0-1>,
615       "patch_fields": {
616         "field": "<generator or ShotIR field to modify>",
617         "value": "<new value, constraint, or reference to insert>"
618       },
619       "rationale": "<short reason based only on scalar scores and specification>"
620     }
621   ],
622   "global_notes": "<one-sentence summary>",
623   "uncertain": <true or false>
624 }
625
626 Rules:
627 1. Output valid JSON only.
628 2. Do not claim visual evidence because frames are not provided.
629 3. Do not output frame IDs or evidence keyframes.
630 4. Use temporal_span = null unless segment-level scalar scores are provided.
631 5. Every patch must use an action from the supported repair-plan vocabulary.
632 6. If the scalar scores do not indicate a clear repair target, return:
633 {"patches": [], "global_notes": "No reliable score-based repair action.", "uncertain": true}

```

634 **F.5 Parsing and invalid-output handling.** Invalid JSON outputs are repaired using a deterministic  
635 parser when possible. If the output cannot be parsed, the prediction is marked invalid. Failure  
636 codes outside the taxonomy are mapped to the nearest taxonomy code only when an exact synonym  
637 exists; otherwise they are counted as false positives. Predictions without temporal spans receive  
638 no localization credit. Hallucinated failures without visual evidence are counted as false positives.  
639 Repair actions outside the supported repair-plan vocabulary are marked invalid for repair-plan scoring  
640 and non-adapter-executable for adapter scoring.

## 641 I Blind Human Repair Evaluation Protocol

642 Raters are shown the specification and one output video, without method names or diagnosis records.  
643 They answer whether the video satisfies the specification overall, whether the target failure is resolved,  
644 whether the repair introduces new artifacts, and how useful the patch is on a 1–5 scale. Human  
645 Pass@ $K$  is the fraction of samples judged successful after at most  $K$  repair attempts.

**Table 33:** Blind human repair evaluation rubric for patch usefulness.

| Score | Criterion                                                                               |
|-------|-----------------------------------------------------------------------------------------|
| 1     | The patch does not address the target failure.                                          |
| 2     | The patch is related to the failure but is not actionable or is largely insufficient.   |
| 3     | The patch is actionable but incomplete or requires substantial manual interpretation.   |
| 4     | The patch directly addresses the target failure with minor ambiguity.                   |
| 5     | The patch directly resolves the target failure and preserves surrounding valid content. |

646 A repaired video is marked as pass if the target failure is resolved and no severe new artifact is  
647 introduced. Human Pass@K is the fraction of samples judged successful after at most  $K$  repair  
648 attempts. New artifacts are marked when raters observe a visible degradation introduced by repair that  
649 was not present in the original generated video. Each repaired video is rated by 3 independent raters.  
650 We report the majority-vote pass/fail label, the mean patch-usefulness score, and the majority-vote  
651 new-artifact label.

**Table 34:** Error analysis on ambiguous and real-normal samples. The table focuses on no-op behavior and wrong-target repair planning. Best values are boldfaced for interpretable comparison columns; the no\_op\_uncertain rate is reported descriptively rather than bold-ranked.

| Method                        | no_op_uncertain rate | Correct no-op rate | Missed necessary repair | Wrong-target patch | Over-confident patch |
|-------------------------------|----------------------|--------------------|-------------------------|--------------------|----------------------|
| Score-only                    | 0.286                | 0.524              | 0.119                   | 0.171              | 0.321                |
| VLM structured report         | 0.219                | 0.476              | <b>0.096</b>            | 0.229              | 0.396                |
| VLM temporal proposal + patch | 0.257                | 0.518              | 0.108                   | 0.200              | 0.350                |
| VideoGenDoctor-diagnosis      | 0.552                | <b>0.738</b>       | 0.142                   | 0.095              | 0.181                |
| VideoGenDoctor-full           | 0.514                | 0.724              | 0.156                   | <b>0.081</b>       | <b>0.164</b>         |

**Table 35:** Qualitative case-study summary. Cases are sampled from controlled, real-failure, and ambiguous stress subsets.

| Case type                          | #Cases | Majority-judged successful / appropriate | Typical outcome                               |
|------------------------------------|--------|------------------------------------------|-----------------------------------------------|
| Successful structured repair plan  | 24     | 18                                       | localized repair plan resolves target failure |
| Ambiguous case with abstention     | 24     | 16                                       | no_op_uncertain avoids over-repair            |
| Over-repair / wrong-target failure | 24     | 7                                        | concrete patch targets wrong object or span   |

## 652 J Bootstrap Confidence Intervals

653 We use 10,000 video-level bootstrap resamples and report percentile 95% confidence intervals. For the  
654 controlled fixture, resampling is performed over 324 videos. For the real-failure subset, resampling is  
655 performed over 240 videos. For blind human repair evaluation, resampling is performed over the  
656 stratified human-rated subset. For paired comparisons, we resample videos with replacement and  
657 compute the distribution of metric differences between two methods. Small point-estimate differences  
658 are not treated as meaningful unless the paired confidence interval excludes zero.

**Table 36:** Leave-one-code-family-out rule ablation. For the held-out family, dedicated family-specific rules are disabled; only generic embedding drift, optical-flow anomaly, scene-change peaks, and structured VLM verification are retained.

| Held-out family | Macro-F1 | tIoU@0.5 | Top-3 hit | Main degradation source                    |
|-----------------|----------|----------|-----------|--------------------------------------------|
| Identity        | 0.782    | 0.579    | 0.536     | face/body-specific anchors removed         |
| Camera          | 0.754    | 0.546    | 0.501     | camera-flow templates disabled             |
| Motion          | 0.721    | 0.512    | 0.468     | jitter / segment-break rules disabled      |
| Alignment       | 0.801    | 0.603    | 0.558     | prop detector and placement rules disabled |
| Style           | 0.836    | 0.641    | 0.590     | compression/style detectors disabled       |
| Scene           | 0.748    | 0.528    | 0.487     | background-anchor rules disabled           |
| Average         | 0.774    | 0.568    | 0.523     | –                                          |

The large degradation under leave-one-code-family-out ablation confirms that VideoGenDoctor should not be interpreted as an unconstrained open-ended failure discovery system. The experiment clarifies which portions of the performance depend on family-specific detectors and which portions remain supported by generic evidence signals and VLM verification.

## K Real-Failure Construction Manifest

**Table 37:** Generator configuration metadata for the real-failure and real-normal subsets. Hosted or version-changing endpoints are logged with access date and endpoint identifier.

| Generator              | Checkpoint / endpoint | fps | Frames | Resolution | Steps | Guidance | Scheduler      | Seed range  | Reference source / prompt rule     |
|------------------------|-----------------------|-----|--------|------------|-------|----------|----------------|-------------|------------------------------------|
| CogVideoX              | CogVideoX1.5-5B       | 16  | 128    | 768×432    | 50    | 6.0      | DPMSolver      | 10000–10059 | ShotIR-to-prompt template v1       |
| Wan                    | Wan2.1-T2V-14B        | 16  | 128    | 768×432    | 50    | 5.0      | FlowMatchEuler | 18000–18059 | ShotIR-to-prompt template v1       |
| Stable Video Diffusion | SVD-XT-1.1            | 14  | 112    | 768×432    | 40    | 7.5      | EulerDiscrete  | 26000–26059 | reference frame from ShotIR anchor |
| HunyuanVideo           | HunyuanVideo-1.5      | 16  | 128    | 768×432    | 50    | 7.0      | FlowMatchEuler | 34000–34059 | multi-shot prompt template v1      |

When a generator does not expose one of these configuration fields, the corresponding manifest value is set to null rather than omitted. For hosted or version-changing endpoints, we log the endpoint identifier, access date, raw request payload, and raw response metadata.

To make the real-failure subset auditable and reproducible, each generated video is accompanied by a JSON manifest. The manifest records the generator identity, checkpoint, input specification, generation configuration, output location, screening decision, verified failure codes, temporal evidence spans, and annotator identifiers. This protocol turns the real-failure subset from an informal collection of generated examples into a traceable evaluation resource.

Each manifest entry contains the following fields: `generator`, `checkpoint`, `prompt_id`, `input_type`, `shotir_spec`, `shotir_to_prompt_converted_text`, `reference_frame_id`, `seed`, `resolution`, `fps`, `num_frames`, `sampling_steps`, `guidance_scale`, `output_path`, `screening_status`, `verified_codes`, `evidence_spans`, and `annotator_ids`. The `screening_status` field takes one of four values: `generated`, `retained_candidate`, `verified`, or `excluded_ambiguous`. Only entries marked as `verified` are included in the reported real-failure evaluation.

```
{
  "video_id": "real_wan_00037",
  "generator": "Wan",
  "checkpoint": "Wan2.1-T2V-14B",
  "prompt_id": "shotir_prompt_037",
  "input_type": "text / ShotIR-to-prompt",
  "shotir_spec": {
    "shot_id": "S037",
    "duration_sec": 8.0,
    "characters": [
      {
        "id": "person_A",
        "description": "young man wearing a blue hoodie",
        "identity_anchor": "ref_person_A_004"
      }
    ],
    "required_props": [
      {
        "id": "red_water_bottle",
        "description": "red water bottle held in the right hand",
        "expected_presence": "visible throughout the shot"
      }
    ],
    "camera": {
      "shot_type": "medium shot",
      "movement": "slow left-to-right tracking",

```

```

705     "viewpoint": "front three-quarter view"
706 },
707 "action_order": [
708     "person_A enters the park path",
709     "person_A jogs while holding the red water bottle",
710     "person_A slows down near the bench"
711 ],
712 "style": {
713     "lighting": "natural daylight",
714     "visual_style": "realistic video"
715 },
716 "scene": {
717     "location": "urban park path",
718     "background_anchor": "trees and bench remain stable"
719 }
720 },
721 "shotir_to_prompt_converted_text": "A realistic 8-second medium shot of a young man wearing a k
722 "reference_frame_id": null,
723 "seed": 18473,
724 "resolution": "768x432",
725 "fps": 16,
726 "num_frames": 128,
727 "sampling_steps": 50,
728 "guidance_scale": 7.5,
729 "output_path": "real_failures/wan/real_wan_00037.mp4",
730 "screening_status": "verified",
731 "verified_codes": [
732     "AL_PROP_MISSING",
733     "CA_MOVE_WRONG",
734     "MO_SEGMENT_BREAK"
735 ],
736 "evidence_spans": [
737     {
738         "failure_code": "AL_PROP_MISSING",
739         "target": "red_water_bottle",
740         "start_sec": 3.25,
741         "end_sec": 6.75,
742         "evidence_keyframes": [
743             "frame_052@3.25s",
744             "frame_080@5.00s",
745             "frame_108@6.75s"
746         ],
747         "annotator_confidence": 0.91,
748         "notes": "The specified red water bottle is not visible after the character begins jogging"
749     },
750     {
751         "failure_code": "CA_MOVE_WRONG",
752         "target": "camera_movement",
753         "start_sec": 0.00,
754         "end_sec": 4.00,
755         "evidence_keyframes": [
756             "frame_000@0.00s",
757             "frame_032@2.00s",
758             "frame_064@4.00s"
759         ],
760         "annotator_confidence": 0.84,
761         "notes": "The camera remains mostly static rather than performing the requested left-to-right"
762     },
763     {

```



```

764     "failure_code": "MO_SEGMENT_BREAK",
765     "target": "temporal_continuity",
766     "start_sec": 5.50,
767     "end_sec": 5.94,
768     "evidence_keyframes": [
769         "frame_088@5.50s",
770         "frame_095@5.94s"
771     ],
772     "annotator_confidence": 0.78,
773     "notes": "A visible discontinuity occurs near the transition to the final action."
774 }
775 ],
776 "annotator_ids": [
777     "ann_02",
778     "ann_05"
779 ]
780 }

```

781 For privacy and release safety, file paths may be anonymized, but the manifest must preserve all fields  
782 required to reproduce the generation configuration, reconstruct the ShotIR-to-prompt conversion, and  
783 audit the verified evidence spans. When a generator does not expose a configuration field such as  
784 guidance\_scale, the field is set to null rather than omitted.

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The checklist is designed to encourage best practices for responsible machine learning research, addressing issues of reproducibility, transparency, research ethics, and societal impact. Do not remove the checklist: **The papers not including the checklist will be desk rejected.** The checklist should follow the references and follow the (optional) supplemental material. The checklist does NOT count towards the page limit.

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Question: Do the main claims made in the abstract and introduction accurately reflect the paper’s contributions and scope?

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Question: Does the paper discuss the limitations of the work performed by the authors?

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- 839 well-specification, asymptotic approximations only holding locally). The authors should
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- 841 would be.
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- 844 implicit assumptions, which should be articulated.
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- 858 penalize honesty concerning limitations.

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860 Question: For each theoretical result, does the paper provide the full set of assumptions and a

861 complete (and correct) proof?

862 Answer: **[TODO]**

863 Justification: **[TODO]**

864 Guidelines:

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- 872 formal proofs provided in appendix or supplemental material.
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875 Question: Does the paper fully disclose all the information needed to reproduce the main

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884 and data are provided or not.
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886 make their results reproducible or verifiable.
- 887 • Depending on the contribution, reproducibility can be accomplished in various ways. For  
888 example, if the contribution is a novel architecture, describing the architecture fully might  
889 suffice, or if the contribution is a specific model and empirical evaluation, it may be necessary  
890 to either make it possible for others to replicate the model with the same dataset, or provide  
891 access to the model. In general, releasing code and data is often one good way to accomplish  
892 this, but reproducibility can also be provided via detailed instructions for how to replicate the  
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894 model checkpoint, or other means that are appropriate to the research performed.
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899 reproduce that algorithm.
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901 architecture clearly and fully.
  - 902 (c) If the contribution is a new model (e.g., a large language model), then there should either  
903 be a way to access this model for reproducing the results or a way to reproduce the model  
904 (e.g., with an open-source dataset or instructions for how to construct the dataset).
  - 905 (d) We recognize that reproducibility may be tricky in some cases, in which case authors are  
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907 closed-source models, it may be that access to the model is limited in some way (e.g.,  
908 to registered users), but it should be possible for other researchers to have some path to  
909 reproducing or verifying the results.

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911 Question: Does the paper provide open access to the data and code, with sufficient instructions to  
912 faithfully reproduce the main experimental results, as described in supplemental material?

913 Answer: [TODO]

914 Justification: [TODO]

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926 method and baselines. If only a subset of experiments are reproducible, they should state  
927 which ones are omitted from the script and why.
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929 (if applicable).
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931 is recommended, but including URLs to data and code is permitted.
- 932
- 933

934 **6. Experimental setting/details**

935 Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters,  
936 how they were chosen, type of optimizer) necessary to understand the results?

937 Answer: **[TODO]**

938 Justification: **[TODO]**

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942 is necessary to appreciate the results and make sense of them.
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944 **7. Experiment statistical significance**

945 Question: Does the paper report error bars suitably and correctly defined or other appropriate  
946 information about the statistical significance of the experiments?

947 Answer: **[TODO]**

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969 **8. Experiments compute resources**

970 Question: For each experiment, does the paper provide sufficient information on the computer  
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974 Justification: **[TODO]**

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